

The optic nerve head as a biomechanical structure: a new paradigm for understanding the role of IOP-related stress and strain in the pathophysiology of glaucomatous optic nerve head damage

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Abstract

We propose here a conceptual framework for understanding the optic nerve head (ONH) as a biomechanical structure. Basic principles of biomechanical engineering are used to propose a central role for intraocular pressure (IOP)-related stress and strain in the physiology of ONH aging and the pathophysiology of glaucomatous damage. Our paradigm suggests that IOP-related stress and strain (1) are substantial within the load-bearing connective tissues of the ONH even at low levels of IOP and (2) underlie both ONH aging and the two central pathophysiologies of glaucomatous damage—mechanical failure of the connective tissues of the lamina cribrosa, scleral canal wall, and peripapillary sclera, and axonal compromise within the lamina cribrosa by a variety of mechanisms. Modeling the ONH as a biomechanical structure generates a group of testable hypotheses regarding the central mechanisms of glaucomatous damage and provides a logic for classifying the principal components of the susceptibility of an individual ONH to a given level of IOP.

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1. Introduction

The clinical hallmarks of a glaucomatous optic neuropathy are glaucomatous excavation of the tissues of the optic nerve head (ONH)—a progressive posterior displacement of the surface of the ONH and progressive excavation of the prelaminar tissues beneath the anteriormost aspect of the scleral canal, the anterior scleral ring (Fig. 1, left) (Ernest and Potts, 1968)—and glaucomatous visual field loss, which most commonly starts as a nasal step and progresses through an arcuate scotoma to full hemifield loss. These clinical hallmarks distinguish a glaucomatous optic neuropathy from the many other optic neuropathies in which damage to the retinal ganglion cell axons, either at the nerve head or within the orbital optic nerve and chiasm, leads to retinal ganglion cell death. Although exceptions exist (Quigley and Anderson, 1976, 1977c; Minckler et al., 1977; Minckler, 1986), ischemic, inflammatory, and compressive damage to the nerve head, orbital optic nerve, or optic chiasm usually results in pallor and atrophy of the nerve head but little or no excavation of the remaining rim tissue (Fig. 1, right). In addition, in these entities the pattern of axon damage within the optic nerve (Anderson and Hendrickson, 1974), and as detected by visual field testing (Anderson and Hendrickson, 1974), is usually different from that of glaucomatous optic neuropathy.

The mechanisms of glaucomatous damage to the tissues of the ONH remain controversial (Fechtner and Weinreb, 1994; Schumer and Podos, 1994). For the past 30 years, discussion has focused on how the retinal ganglion cell axons are damaged within the lamina cribrosa, and controversy has centered on whether intraocular pressure (IOP) (the mechanical hypothesis) or ONH blood supply (the vascular hypothesis) is responsible for ONH axonal damage in this disease.

Like other investigators (Van Buskirk and Cioffi, 1992, 1993; Van Buskirk, 1994), we believe that separating the effects of IOP from the effects of ONH blood flow in the pathophysiology of glaucomatous damage to the ONH is artificial. In place of the “mechanical vs. vascular” paradigm (which concentrates on the mechanism of axonal insult only), we

propose the larger concept of the ONH as a biomechanical structure. This paradigm suggests that the influence of IOP-related connective tissue stress (force/cross sectional area) and strain (local deformation) is central to both the physiology and pathophysiology of all the three ONH tissue types—(1) connective (load-bearing connective tissues of the peripapillary sclera, scleral canal wall, and lamina cribrosa), (2) axonal (retinal ganglion cell axons), and (3) cellular (astrocytes, glial cells, endothelial cells, and pericytes, along with their basement membranes)—as well as the volume flow of blood that nourishes them.

Within this paradigm, it is hypothesized that the ONH connective tissues are exposed to substantial levels of IOP-related stress/strain at normal levels of IOP, which further increase as IOP is elevated (Fig. 2) (Greene, 1980; Bellezza et al., 2000). Stresses and strains at a given level of IOP are physiologic or pathophysiologic depending upon the response of the tissues that experience them (Fig. 3). Physiologic stress/strain induces a broad spectrum of changes in both the connective tissues and blood vessels that are central to normal aging. Pathophysiologic stress/strain induces pathologic changes in cell synthesis and tissue microarchitecture (Fig. 3) that underlie the two governing pathophysiologies in glaucoma: (1) mechanical failure of the load-bearing connective tissues of the ONH, and (2) progressive damage to the adjacent axons by a variety of mechanisms (Fig. 4).

The paradigm proposes that the two pathophysiologies of glaucoma develop and progress independently, albeit with interactions, during each stage of the neuropathy (Fig. 4). However, the progression of the neuropathy within an individual ONH is very much influenced by the physiologic age of the tissues in which it occurs.

Thus, although the levels of IOP-related stress and strain are central to the principal physiologic and pathophysiologic processes (Fig. 3) within each tissue, it is the interplay between (1) the resultant connective tissue damage, (2) axonal compromise, and (3) the physiologic age of the tissues (Fig. 4) that determines how a particular ONH comes to look glaucomatous.

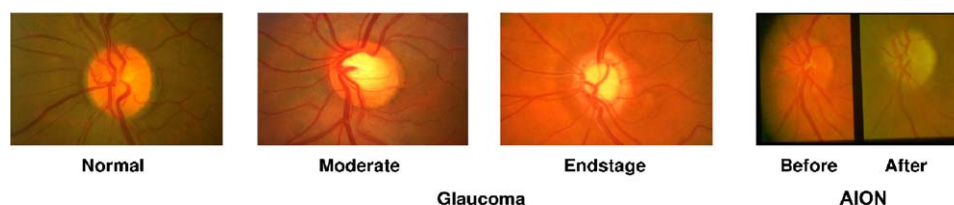


Fig. 1. Progressive deformation and excavation of the tissues of the ONH beneath the anterior rim of the scleral canal is the clinical hallmark of a glaucomatous optic neuropathy (left). Axonal damage resulting from an episode of AION, right, usually results only in ONH pallor. AION images reprinted from Quigley, H.A., Anderson, D.R., 1977. Cupping of the optic disc in ischemic optic neuropathy. *Trans. Am. Acad. Ophthalmol. Otolaryngol.* 83, 755–762, Copyright, 1977, with permission from *American Academy of Ophthalmology*.

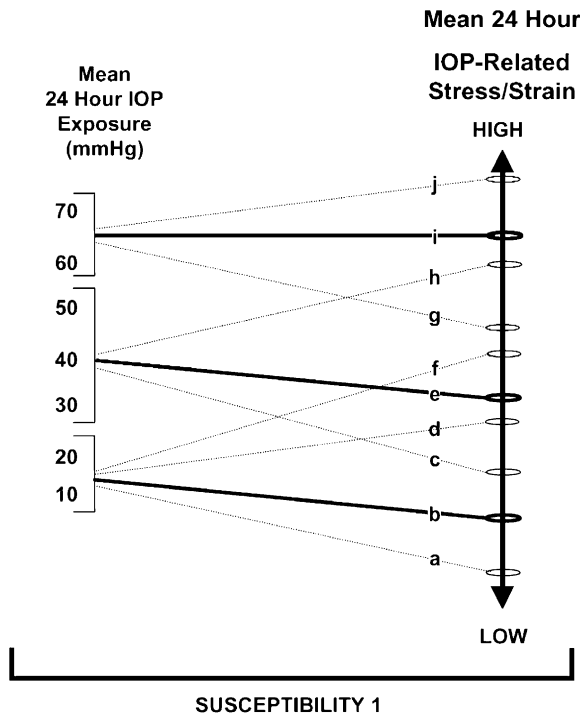


Fig. 2. The relationship between IOP and IOP-related stress (susceptibility 1). The magnitude of IOP-related stress for a given level of IOP is principally determined by the 3D anatomy of the eye, the scleral shell, the lamina cribrosa, and the scleral canal, and is the first component of ONH susceptibility to IOP-related damage. For most ONHs, IOPs of 6–21 mm Hg generate low levels of IOP-related stress, which are still substantial (solid line b, dotted line d). However, some ONHs will have unusually low (line a) or high (line f) levels of IOP-related stress based on their 3D geometry even at these low levels of IOP. IOPs of 22 mm Hg to the high 30s create higher levels of IOP-related stress for most patients (solid line e, dotted line h), but some patients may experience lower (dotted line c) levels of stress at these IOPs. IOPs above the high 30s create very high levels of IOP-related stress for the overwhelming majority of patients (solid line i, dotted line j), but a few may still experience lower levels of stress even at such high IOP (dotted line g). Whether a given level of IOP-related stress creates physiologic or pathophysiologic effects within the ONH tissues is a separate component of susceptibility with its own determinants, as seen in Fig. 3.

Whereas the ONH is the primary site of damage for a group of optic neuropathies, only a subset of these disorders assume the clinical appearance and behavior commonly associated with the term *glaucomatous*. What then constitutes a glaucomatous ONH? We propose that the defining phenomenon that underlies the glaucomatous optic neuropathies is damage to the neural and connective tissues of the ONH, which is governed by the distribution of IOP-related connective tissue stress and strain, regardless of the mechanism of insult or the level of IOP at which that insult occurs. Said in another way, *glaucomatous* is the term used to describe the clinical appearance and behavior the ONH assumes as its neural and connective tissues are damaged (1) in a pattern and

(2) by the several pathophysiologic processes governed by IOP-related connective tissue stress and strain.

In this article, we present a detailed explanation of our concept of the ONH as a biomechanical structure, review initial experimental evidence in support of these concepts, and discuss the principal clinical and research implications thereof.

2. The ONH as a biomechanical structure

When we say that the ONH is a biomechanical structure, we mean that the conditions of its mechanical behavior can be described and predicted from its geometry, material properties, boundary conditions, and mechanical loading. In a simple analysis, the mechanical loading of the ONH is principally the result of IOP, which generates mechanical stress (force/cross-sectional area) and mechanical strain (a measure of local deformation) within its load-bearing connective tissues. IOP-related stress is substantial within the ONH connective tissues even when IOP is low due to the concentration of stress around any hole (the scleral canal) in a pressurized spherical shell (the posterior scleral shell) (Timoshenko and Goodier, 1970; Bellezza et al., 2000).

Depending on the material properties of the connective tissues, mechanical stress may result in connective tissue deformation or strain. Only strain, or physical deformation of the tissue, is capable of inducing a physiologic stress within the adjacent axons, astrocytes, and endothelial cells. (Here, *physiologic stress* is defined as physical and metabolic changes within a cell in response to alterations in its environment.) Thus IOP-related stress, which is dependent upon one set of risk factors, may generate IOP-related strain, which is determined by the material properties of the connective tissues (an additional risk factor). Only IOP-related strain can affect the surrounding tissues and it does so by a variety of mechanisms.

The biomechanical notion that all of the ONH tissues respond to the presence of IOP-related strain within the load-bearing connective tissues evolves from seminal experiments that established the following facts: (1) Axonal transport is compromised at the level of the lamina cribrosa at physiologic levels of IOP (Ernest and Potts, 1968; Minckler, 1986) and is further impaired following acute (Quigley and Anderson, 1976, 1977c; Minckler et al., 1977; Minckler, 1986) and chronic (Anderson and Hendrickson, 1974; Quigley and Anderson, 1976, 1977c; Minckler et al., 1977) IOP elevations. (2) ONH, retinal, and choroidal blood flow are all affected in different ways by acute (Alm and Bill, 1970, 1972; Alm, 1973; Geijer and Bill, 1979; Sossi and Anderson, 1983; Weinstein et al., 1983) and chronic (Grunwald et al., 1984; Pillunat et al., 1985, 1987; Ulrich

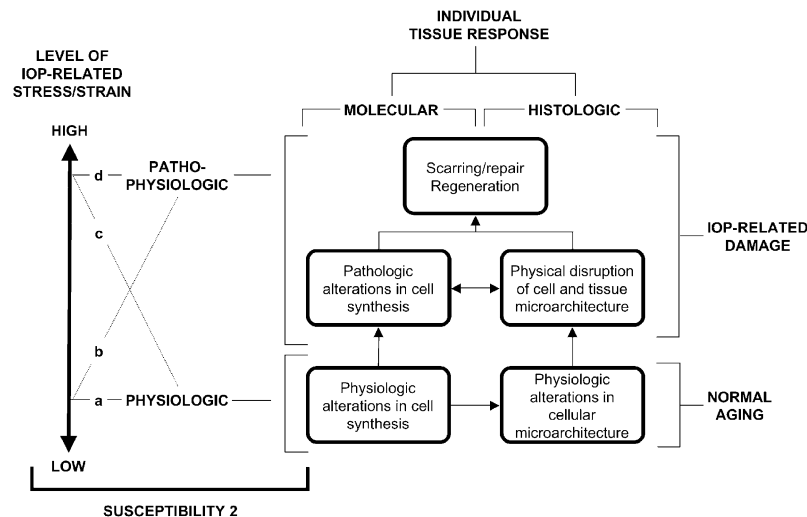


Fig. 3. A given level of IOP-related stress is physiologic or pathophysiologic for a given ONH based on the effects it induces within each ONH tissue type (Susceptibility 2). For most ONHs, low levels of IOP-related stress can be expected to induce physiologic changes within the tissues of the ONH (line a). Physiologic effects underlie normal tissue aging and include alterations in cell synthesis at the molecular level, leading to alterations in intracellular and extracellular microarchitecture that can be seen by light and electron microscopy. However, some ONHs are susceptible to even low levels of IOP-related stress, and thus their tissues would be found to demonstrate pathophysiologic responses (line b). Pathophysiologic effects mark the onset of IOP-related damage and progress from pathologic changes in cell synthesis to histologic evidence of frank disruption of the cellular microarchitecture with eventual scarring, repair, and perhaps, in the case of the axons, regeneration. Most ONHs are susceptible to high levels of IOP-related stress (line d). However, some ONHs can be expected to be more resilient (line c).

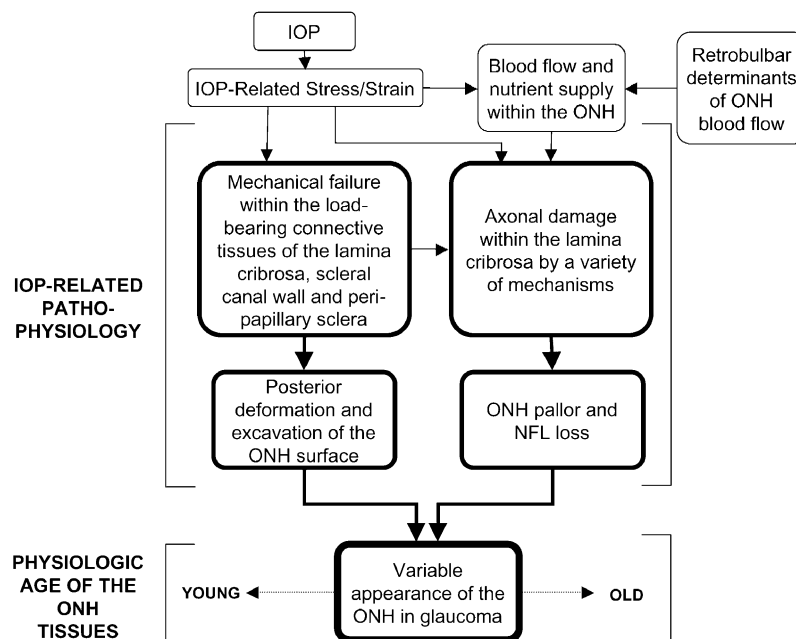


Fig. 4. Two separate but interactive pathophysiologies underlie IOP-related damage to the tissues of the ONH and they are the principal determinants of its appearance. A predictable pattern of mechanical failure within the load-bearing connective tissues of the ONH leads to the classic posterior deformation and excavation of the glaucomatous ONH (see also Fig. 7). Axons are separately damaged within the lamina by a variety of mechanisms and their loss leads to ONH pallor. Note that the pathophysiology of ganglion cell death (either primary or secondary to axonal trauma within the lamina cribrosa) may underlie a third level of susceptibility to IOP (not shown), but this pathophysiology should contribute only pallor and nerve fiber layer dropout to the appearance of the ONH. Finally, these two pathophysiologies can be expected to have markedly different effects in young vs. old ONHs.

et al., 1986) IOP elevations. (3) A predictable pattern of connective tissue deformation underlies glaucomatous cupping (Emery et al., 1974; Quigley et al., 1983). (4) A

predictable pattern of axonal loss underlies glaucomatous visual field loss (Quigley and Green, 1979; Quigley et al., 1982a). (5) The ONH surface (Quigley, 1977;

Zeimer et al., 1981; Pederson and Herschler, 1982; Zeimer and Ogura, 1989; Coleman et al., 1991; Burgoyne et al., 1994, 1995a; Quigley and Pease, 1996; Park and Hong, 1998) and lamina cribrosa (Levy and Crapps, 1984; Yan et al., 1994) are compliant structures, bowing posteriorly and returning to baseline following acute elevations of IOP. (*Compliance* is here defined as the ease of deformation in response to an applied load.) (6) ONH surface hypercompliance (defined as a pathologically exaggerated ease of deformation in response to an applied load) occurs early in experimental glaucoma (Burgoyne et al., 1995b), which suggests that damage to the connective tissues occurs early in the pathophysiology of IOP-related damage to the ONH. (7) Profound changes in the extracellular matrix of the load-bearing tissues occur in aging (Hernandez et al., 1989; Repka and Quigley, 1989) and in glaucoma (Minckler and Spaeth, 1981; Tengroth and Ammitzboll, 1984; Hernandez et al., 1990; Morrison et al., 1990), and IOP-induced alterations in the synthetic activities of the cells associated with these tissues may underlie these changes (Hernandez et al., 1994a, b; Clark et al., 1995).

3. The relationship between IOP and IOP-related stress

The connective tissues of the peripapillary sclera, the lamina cribrosa, and the scleral canal wall bear the forces generated by the IOP and are thus the load-bearing tissues of the ONH (Zeimer, 1995). For a given ONH, IOP-related force has a predictable distribution and leads to predictable levels of IOP-related stress. It is fundamental to our concept that IOP-related stress within the connective tissues of the ONH is not only substantial even at low levels of IOP, but is also increased as IOP increases. We hypothesize that (1) the magnitude and distribution of IOP-related stress within a given ONH for a given level of IOP is primarily determined by the three-dimensional (3D) anatomy of the eye, the scleral shell, the lamina cribrosa, and the scleral canal (Greene, 1980; Bellezza et al., 2000); and (2) the mechanical response of the load-bearing tissues to a given level of IOP-related stress (strain) is determined by their material properties (Zeimer, 1995).

To begin the study of the relationship between IOP, IOP-related stress, and IOP-related strain within the connective tissues of the ONH (Yan et al., 1994; Dongqi and Zeqin, 1999; Bellezza et al., 2000), we are building finite element models of the scleral canal as a hole in a pressurized spherical shell. To our knowledge, Greene (1980) was the first to predict that IOP-related stress within the peripapillary sclera would be further magnified due to the concentration of stress around any hole in a pressurized spherical shell.

Our initial finite element models (Bellezza et al., 2000) demonstrated IOP-generated stress of approximately 10–17 times IOP within the tissues of the scleral shell at points distant from the ONH, 30 times IOP within the connective tissues of the peripapillary sclera, 30–100 times IOP within the scleral canal wall, and 50–180 times IOP within the laminar beams, depending on the 3D geometry of these tissues.

4. The relationship between stress, strain, material properties, and mechanical behavior

The extracellular matrix of the sclera and lamina cribrosa resists tensile, compressive, and shear stresses (Morrison, 1995; Zeimer, 1995; Hernandez and Gong, 1996). (Tensile stress generates elongation of an object along the applied loading axis; compressive stress generates compression of an object along the applied loading axis; shear stress skews the shape of an object by pulling parallel planes either in opposite directions or by different relative magnitudes). Within the laminar beams, fibrils of collagen types I and III and elastin fibers are longitudinally oriented within a dense proteoglycan matrix (Hernandez et al., 1987; Morrison et al., 1988, 1989b; Hernandez, 1992). At the insertion of the peripheral laminar beams into the scleral canal wall (the laminar insertion zone), elastin fibers from the beams insert into a circle of elastin that rings the scleral canal (Quigley et al., 1991a, b; Hernandez, 1992).

Mechanical testing of soft tissues has demonstrated that collagen and elastin (but principally collagen) resist tensile stress (Vogel, 1974) and the proteoglycan matrix resists compressive stress (Larrabee, 1986). Collagen fibers first straighten (substantially) and then elongate (minimally) to resist a uniaxially applied load (load applied in a single direction) (Kenedi et al., 1965; Broom, 1978). Tensile stiffness and strength are hypothesized to be manifestations of collagen cross-linking and fiber diameter, as well as interactions between collagen and the glycosaminoglycans (GAGs) of the extracellular matrix (Vogel, 1980; Oxlund and Andreassen, 1980). Tissue elasticity, defined as resumption of initial configuration after removal of an applied load, is due to collagen and elastin (Oxlund et al., 1988). Thus, the composition of the extracellular matrix, in terms of relative amounts and types of collagen, elastin, GAGs, and other constituents, determines the material properties of a tissue and, consequently, its behavior in response to applied force.

Consider a steel cable of a given cross-sectional area suspended tautly between two supports (Fig. 5A). The forces generated within the cable for a given load are determined only by the magnitude of the load. However, the stress (force/cross-sectional area) generated within the cable for a given load is determined by

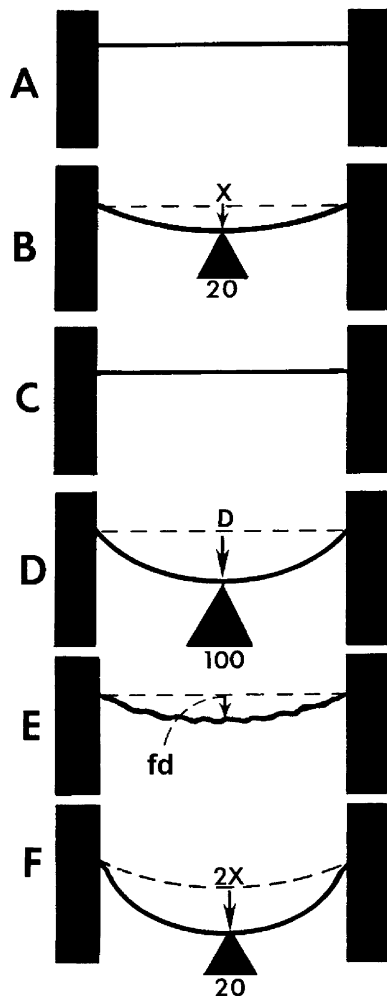


Fig. 5. This diagram represents the relationship between force, stress, material properties, and mechanical behavior. A steel cable fixed between two supports (A) is deformed an amount, X , by a load of 20 lbs (B). The forces generated within the cable (force/cross-sectional area) for a given load are determined only by the magnitude of the load. However, the stresses generated within the cable (force/cross-sectional area) for a given load are determined by the cross-sectional area of the cable. The mechanical behavior of the cable is dictated by the cable's material properties. The 20-lb load is within the elastic limits of the cable because, on release, the cable resumes its original (pre-load) configuration (C). The 100-lb load (D), however, exceeds the elastic limits of the cable, resulting in permanent or "fixed" deformation (fd) after the load is removed (E) and an excessive amount of deformation (hypercompliance) for the same 20-lb load (F).

the cross-sectional area of the cable. Thus, a thin cable experiences a higher level of stress than a thick cable for the same amount of load.

The mechanical behavior of the cable, at a given level of stress, is measured by deformation and is dictated by the cable's material properties. (By mechanical behavior we mean all aspects of the object's response to an applied stress). One aspect of this behavior is the cable's strain (local deformation in response to an applied stress) and one measure of strain is how much the cable

deforms for a given vertical load. In our example, the deformation is represented by X for 20 lbs (Fig. 5B) and D for 100 lbs (Fig. 5D). The 20-lb load is within the elastic limits of the cable's material properties if, upon release, the cable resumes its original (pre-load) configuration (Fig. 5C). The 100-lb load, however, damages the cable, resulting in permanent deformation (Fig. 5E) and hypercompliance (Fig. 5F: $2X$ deformation for the same 20-lb load).

Mechanical failure, then, is the process by which strain-induced damage to an object (or tissue) alters its ability to resist force. Permanent deformation and hypercompliance (Burgoyne et al., 1995b) are changes in the macroscopic behavior of a tissue that follow the onset of mechanical failure.

5. The effects of IOP-related connective tissue strain on the connective tissues, astrocytes, and axons depend on the state of the connective tissues

We propose that IOP-related strain has different effects on the three principal tissues of the ONH depending on the state of the load-bearing connective tissues. When the load-bearing connective tissues are undamaged and still within their elastic limits, individual connective tissue trabeculae may be deformed posteriorly, depending on the level of IOP-related stress and the material properties of the tissue, but if the stress is reduced, the deformation vanishes, implying that the extracellular matrix of the tissues has not been damaged. It is important to recognize that even though the ONH connective tissues are intact, IOP-related damage to the adjacent axons and living cells can still occur by a variety of mechanisms (see below). Thus, IOP-related strain can generate pathophysiologic responses within the adjacent axons and living cells at levels at which the connective tissues themselves remain intact.

Once the connective tissues are loaded beyond their elastic limits and the extracellular matrix is damaged, the pathophysiologic effects of connective tissue damage (such as basement membrane disruption, astrocyte and glial cell activation, etc.) are added to the effects of the underlying distribution of IOP-related strain. It is of profound importance to understand that once the load-bearing connective tissues are damaged, further damage to the connective tissues themselves and damage to the adjacent axons and living cells can be expected to occur, by whatever mechanism, even at lowered levels of IOP. It is also important to recognize that the effects of IOP-related strain on the ONH connective tissues (damaged or undamaged) may be further confounded by the molecular factors expressed by the constituent laminar astrocytes and glia if, in fact, these cells are activated by axonal distress (Burgoyne and Morrison, 2001) separate from laminar strain.

5.1. Load-bearing connective tissues

5.1.1. Connective tissues undamaged (within their elastic limits)

A variety of biological tissues have been shown to undergo synthesis of extracellular matrix in response to a change in applied load (Schneider and Deck, 1981; Ito et al., 1987; Koob and Vogel, 1987; Zamora and Marini, 1988; Prosser et al., 1989; Tumminia et al., 1998). Within the ONH, the laminar astrocytes have been shown to respond to changes in IOP-related stress (Hernandez, 1993; Hernandez et al., 1994a; Pena et al., 1996; Varela and Hernandez, 1997) and, in culture, to acute changes in barometric pressure (Yang et al., 1993).

Within individual laminar trabeculae, levels of tensile strain, either physiologically fluctuating or pathologically elevated, that are within the elastic limits of the tissue may still (1) induce stiffening of the connective tissues by the laminar astrocytes (mechanisms may include an increase in the amount and/or a change in the type of collagen, and a change in hydration levels and types of GAGs), (2) induce basement membrane thickening by both the astrocyte and endothelial cells, and (3) secondarily diminish axonal nutrient diffusion from the laminar capillaries to the adjacent axon bundles, as a result of the changes to the extracellular matrix and basement membranes described in (1) and (2) (Fig. 6).

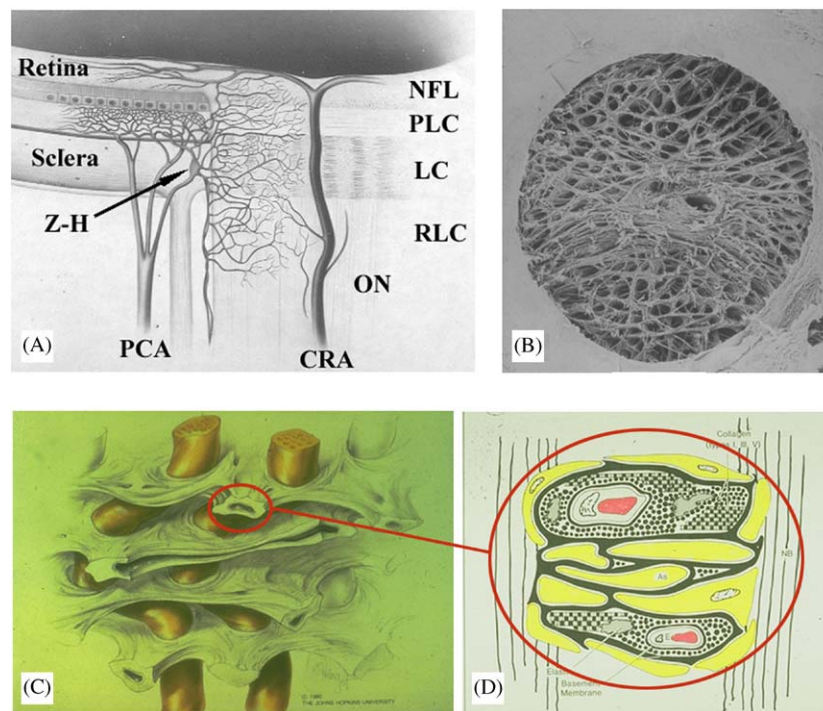


Fig. 6. IOP-related stress may have acute and chronic effects on the delivery of blood-borne nutrients to the axons within the lamina cribrosa. (A) The blood supply to the ONH. The traditional extra-bulbar determinants of ONH laminar capillary volume flow are fluctuations in systemic blood pressure and vasospasm. However, laminar volume flow may additionally be diminished by a compressive effect on volume flow within the branches of the posterior ciliary arteries that penetrate to the choroid, prelaminar, laminar, and post-laminar regions from IOP-related stress within the peripapillary sclera. Separate from these effects, IOP-related stress within each individual laminar trabecula (B and C) may have acute compressive effects on laminar capillary volume flow (solid red, D). Separate from considerations of volume flow, axonal nutrition within the lamina (D) requires diffusion of nutrients from the laminar capillaries (solid red), across the endothelial and pericyte basement membranes, through the extracellular matrix of the laminar beam (stippled), across the basement membranes of the astrocytes (thick black), into the astrocytes (yellow), and across their processes (not shown) to the adjacent axons (vertical lines). Chronic age-related changes in the endothelial cell and astrocyte basement membranes, as well as IOP-induced changes in the laminar extracellular matrix and astrocyte basement membranes, may diminish nutrient diffusion to the axons in the presence of a stable level of laminar capillary volume flow. Z-H = circle of Zinn-Haller; PCA = posterior ciliary arteries; NFL = nerve fiber layer; PLC = prelaminar region; LC = lamina cribrosa; RLC = retrolaminar region; ON = optic nerve; CRA = central retinal artery. (A) Reprinted with permission from Cioffi, G.A., Van Buskirk, E.M., 1996. Vasculature of the optic nerve and peripapillary choroid. Chapter 8. In: Ritch, R., Shields, M.B., Krupin, T. (Eds.), *The Glaucomas*, Second ed. Mosby-Year Book, St. Louis. (B) Reprinted with permission from Quigley, H.A., Brown, A.E., Morrison, J.D., Drance, S.M., 1990. The size and shape of the optic disc in normal human eyes. *Arch. Ophthalmol.* 108, 51–57, Copyright, 1990, American Medical Association. (C) Reprinted with permission from Quigley, H.A., 1995. Overview and introduction to session on connective tissue of the optic nerve in glaucoma. Chapter 2. In: Drance, S.M., Anderson, D.R. (Eds.) *Optic Nerve in Glaucoma*. Kugler Publications, Amsterdam/New York. (D) Reprinted with permission from Morrison, J.C., L'Hernault, N.L., Jerdan, J.A., Quigley, H.A., 1989. Ultrastructural location of extracellular matrix components in the optic nerve head. *Arch. Ophthalmol.* 107, 123–129, Copyright, 1989, American Medical Association.

5.1.2. Connective tissues damaged (undergoing some phase of mechanical failure)

The histology of mechanical failure within soft tissues is controversial. [Torp et al. \(1974a,b\)](#) mechanically tested rat tail tendon and described the histologic features associated with the initial toe region (straightening of collagen), linear region (lengthening of collagen fibrils), and yield and failure region (progressive dissociation of the collagen fibrils) of the stress–strain curve. [Craik and McNeil \(1965\)](#) described changes in the trichrome staining pattern of collagen (from green to red) and the onset of fracture lines within collagen seen by light microscopy in human skin stretched to failure. [Quigley et al. \(1991b\)](#) reported disruption of both collagen and elastin in the laminar insertion sites of human and monkey ONHs with early glaucomatous damage. They hypothesized that the disruption of collagen may have preceded the change in elastin.

Based on these observations, we hypothesize that IOP-related disruption of collagen is the earliest stage of connective tissue damage in glaucoma and precedes the damage to elastin, disruption of the lamina, and remodeling of the extracellular matrix that have been well characterized ([Quigley et al., 1981, 1991a,b](#); [Morrison et al., 1989a, 1990](#); [Hernandez et al., 1990](#); [Hernandez, 1992](#)). Regardless of the specific inciting event, strain levels that exceed the elastic limits of a soft tissue should lead to disruption of the tissues, verifiable by light or electron microscopy, which, in turn, leads to scarring and repair ([Fig. 3](#)).

5.2. Retinal ganglion cell axons

5.2.1. Connective tissues undamaged

IOP-related damage to the axons that occurs while the connective tissues are within their elastic limits can result from a variety of mechanisms ([Fig. 4](#)). The principal effect should be ONH pallor due to axon loss, without excavation of the ONH surface, because, by definition, the connective tissues are within their elastic limits and still able to resist IOP-related load. The principal mechanisms may include (1) axonal ischemia through a variety of mechanisms (the traditional vascular hypothesis ([Fechtner and Weinreb, 1994](#))), (2) physical compression of the axons secondary to the deformation of intact laminar trabeculae (the traditional mechanical hypothesis ([Fechtner and Weinreb, 1994](#))), and (3) spontaneous axonal compression secondary to tissue pressure differences across the intact lamina cribrosa (Yablonski's theory of spontaneous axonal compression ([Yablonski and Asamoto, 1993](#))). Regardless of the mechanism of insult, the astrocytes and glia may mediate its effects ([Burgoyne et al., 2001](#)).

The traditional mechanical hypothesis of axonal damage and Yablonski's theory of spontaneous axonal compression have been well described elsewhere

([Yablonski and Asamoto, 1993](#); [Fechtner and Weinreb, 1994](#)) and so will not be reviewed here. We will focus instead on a variant of the traditional vascular hypothesis that suggests that ischemic damage to the axons can result from levels of IOP-related strain that are within the elastic limits of the load-bearing tissues.

It has been a basic assumption that within the lamina cribrosa, axonal viability requires the adequate delivery of nutrients, in terms of laminar capillary volume flow, and sufficient diffusion of these nutrients from the laminar capillaries across the endothelial cell basement membranes, through the trabecular extracellular matrix, and across the astrocyte basement membranes to the center of the axon bundles ([Fig. 6](#)). We hypothesize that within the lamina, connective tissue strain has both direct and indirect effects on axonal nutrition, and that axonal ischemia can be the result of either IOP-induced occlusion of the laminar capillaries (direct effect) or decreased diffusion of nutrients (indirect effect), or both.

In terms of only the retrolaminar effects on blood flow, the level of IOP-related strain within the peripapillary sclera may significantly affect volume flow through the scleral branches of the short posterior ciliary arteries ([Langham, 1980](#)) ([Fig. 6](#)). Thus, volume flow through the circle of Zinn Haller and the small penetrating vessels that pass anteriorly to the prelaminar nerve, transversely into the laminar insertion sites, and posteriorly to the pial branches supplying the retrolaminar optic nerve, can be theoretically diminished, individually or collectively, by elevated levels of IOP-related strain ([Langham, 1980](#)).

Separate from the retrolaminar effects, frank occlusion of the capillaries running within each laminar beam may occur due to tensile, compressive, or shear strains within the beam that are still within the beam's elastic limits. This may be more likely to occur when capillary perfusion pressure has been adversely affected by retrolaminar effects of IOP-related strain or non-IOP-related factors such as nocturnal hypotension, diabetes, and retrolaminar vasospasm ([Spaeth, 1977](#); [Langham, 1980](#); [Hayreh et al., 1994](#)). In addition to the effects of diminished volume flow within the laminar capillaries, axonal nutrition may be further impaired as a result of diminished nutrient diffusion from the laminar capillaries to the center of the axon bundles caused by IOP-related thickening of the capillary and astrocyte basement membranes and stiffening of the laminar extracellular matrix ([Farquhar, 1991](#); [Hernandez and Gong, 1996](#)), which would be expected to alter its diffusion properties.

5.2.2. Connective tissues damaged

Once the load-bearing connective tissues are physically damaged, secondary IOP-related damage to axons would be expected to occur at lower levels of IOP by all of the mechanisms described above.

5.3. Astrocytes and endothelial cells

5.3.1. Connective tissues undamaged

Elevated levels of IOP-related strain can be expected to induce synthesis of extracellular matrix by the laminar astrocytes, as outlined above, which alters the material properties of the adjacent laminar trabeculae. Additionally, the laminar astrocytes and laminar capillary endothelial cells may thicken their basement membranes as a secondary response to the physical distortion induced by laminar deformation. Thickening of the basement membranes of the laminar capillary endothelial cells in response to elevated levels of IOP-related stress was reported in a poster by Hernandez's group at the 1997 ARVO meeting (Pena et al., 1997). Basement membranes of the laminar trabeculae have also been noted to thicken with age (Hernandez et al., 1989).

5.3.2. Connective tissues damaged

Laminar trabecular basement membrane thickening is a common finding in published descriptions of the pathology of glaucomatous damage to the ONH tissues (Hernandez et al., 1990; Morrison et al., 1990). Physical disruption of both astrocyte and endothelial basement membranes may be an important stimulus to the onset of scarring and repair within the laminar trabeculae.

6. Effects of IOP-related connective tissue strain on the volume flow of blood and nutrient diffusion

Before it reaches the lamina cribrosa, the volume flow of blood at a given level of IOP is principally influenced by systemic blood pressure and vasospasm. Additionally, it has been hypothesized that IOP-related strain within the peripapillary sclera exerts a compressive effect on volume flow within the branches of the posterior ciliary arteries that penetrate to the choroid, prelaminar, laminar, and post-laminar regions (Langham, 1980).

Separate from these effects, IOP-related strain within each individual laminar trabecula may have acute compressive effects on laminar capillary volume flow. Axonal nutrition within the lamina requires diffusion of nutrients from the laminar capillaries, across the endothelial and pericyte basement membranes, through the extracellular matrix of the laminar beam, across the basement membranes of the astrocytes, into the astrocytes, and across their processes to the adjacent axons. Chronic age-related changes in the endothelial cell and astrocyte basement membranes, as well as IOP-induced changes in the laminar extracellular matrix and astrocyte basement membranes, may diminish nutrient diffusion to the axons even in the presence of a stable level of laminar capillary volume flow.

7. Two principal pathophysiologies underlie IOP-related damage to the ONH

7.1. Mechanical failure within the load-bearing connective tissues

Quigley et al. (1981, 1983) and Jonas et al. (2003) have characterized the backward bowing and compression of the laminar plates that underlie the onset and progression of glaucomatous cupping. Our paradigm suggests that the central pathophysiology within the connective tissues of a glaucomatous ONH is a predictable pattern of physical damage from mechanical failure (Fig. 7). It is the pattern of this damage, as well as the tissue deformation it allows, that underlies the unique appearance of a glaucomatous ONH, whether the damage occurs at low or high IOP.

A central tenet of structural engineering is that the distribution of stress within any load-bearing structure is ultimately predictable and that the structure will deform according to its constituent material properties and fail in a predictable manner based on the distribution of strain. In our scenario (Fig. 7), individual connective tissue trabeculae of the anterior lamina cribrosa mechanically fail, thus transferring the force they were resisting to the immediately adjacent trabeculae and increasing their load for the same level of IOP. Even under a constant level of IOP, the adjacent laminar trabeculae then progressively fail as the same level of overall IOP-induced load is spread over a continually decreasing cross-sectional area of connective tissue. The more trabeculae that fail, the higher the stress on the remaining trabeculae. Thus, in glaucoma, progressive cupping can occur at unchanged or even reduced levels of IOP, if IOP has not been adequately lowered and sometimes even after it has been maximally lowered.

We propose that, at any level of IOP at which early damage to the load-bearing tissues occurs, the lamina cribrosa and its individual constituent trabeculae manifest a change in mechanical behavior that is, itself, macroscopic evidence of damage. Considering the lamina cribrosa as a whole, we hypothesize that the permanent deformation and hypercompliance of the ONH surface (Burgoyne et al., 1995b) and lamina (Bellezza et al., 2003b) demonstrated previously in experimental glaucoma are manifestations of early connective tissue damage.

7.2. Damage to the axons by a variety of mechanisms

As noted above, the biomechanical paradigm suggests that IOP-related damage to the axons can occur by a variety of mechanisms even when the adjacent connective tissues of the lamina cribrosa are intact. It is important to recognize that the traditional discussion of vascular vs. mechanical mechanisms of axonal damage

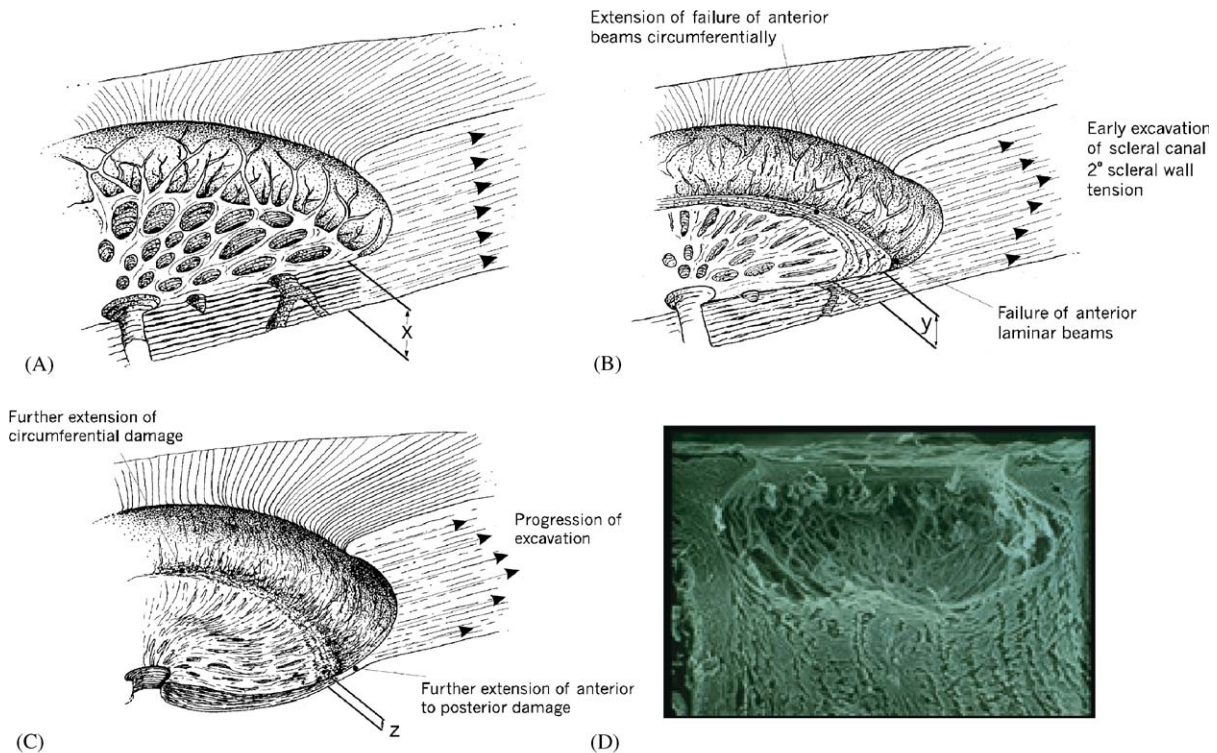


Fig. 7. A predictable pattern of mechanical failure within the load-bearing connective tissues of the ONH underlies the onset and progression of posterior deformation and excavation. (A) Schematic of a normal sagittal view of scleral canal and laminar plates with normal laminar thickness (X) and directions of scleral wall tension (arrows). (B) Early failure of anterior laminar plates causes thinning ($Y < X$), excavation, and circumferential extension, which progresses in (C) ($Z < Y$). Note that it is the mechanical failure of the peripheral laminar beams that allows (1) the remaining lamina to be pushed posteriorly under the direct anterior/posterior effects of IOP (posterior deformation), and (2) the remaining scleral wall tissues to be pulled under the anterior scleral rim owing to the effects of scleral wall tension, recognized clinically as excavation. (D) Further thinning of the lamina and pronounced excavation in an end-stage ONH. (D) Courtesy of Harry A. Quigley, MD.

implied that IOP does not play an important role in vascular or ischemic mechanisms of damage. By contrast, the biomechanical paradigm argues that what defines glaucomatous optic neuropathy is that the location of axonal damage is governed by the distribution of IOP-related stress and strain. Thus, although the paradigm is compatible with the notion that there are multiple mechanisms, it proposes that regardless of mechanism or the level of IOP, the location and pattern of glaucomatous axonal damage is governed by the distribution of IOP-related stress and strain.

The paradigm assumes that the principal site of IOP-related damage to the retinal ganglion cell axons is within the lamina cribrosa (Anderson and Hendrickson, 1974; Minckler et al., 1977; Quigley and Anderson, 1976, 1977c). However, it acknowledges the potential importance of separate pathophysiologies within the lateral geniculate and visual cortex (Yucel et al., 2000, 2001, 2003; Gupta and Yucel, 2001) as well as evidence both for (Kendell et al., 1995; Wygnanski et al., 1995) and against (Panda and Jonas, 1992; Janssen et al., 1997) direct, IOP-induced damage to the retinal photoreceptors. In fact, our conceptual framework is compa-

tible with the notion of additional direct photoreceptor (or other retinal cell) effects, but suggests that (1) in most eyes, for a given level of IOP, the axons within the lamina are likely to be more vulnerable to direct effects of IOP than are the cells within the retina, and (2) direct effects within the retina may contribute to ONH pallor but not to true glaucomatous excavation as described in Figs. 1 and 7.

Retrolaminar (Hayreh, 1974) and prelaminar (Hayreh, 1986) ischemia are the central pathophysiologies of the traditional vascular hypothesis. Based on the general assumption that prelaminar and retrolaminar ONH blood supply is IOP-independent, i.e., principally influenced by factors other than IOP, Hayreh (1974, 1986) proposed that if a glaucomatous form of optic neuropathy could occur due to ischemia in these regions, then this form of optic neuropathy was most likely unrelated to IOP.

Our model is compatible with the notion that prelaminar ischemia, which may be less IOP-dependent than laminar ischemia, may occur in addition to the IOP-dependent pathophysiology within the lamina cribrosa, and in some eyes may predominate. However,

as pointed out above, prelaminar ischemia alone, whether or not it is IOP-induced, should principally cause pallor of the ONH tissues and not glaucomatous excavation. Therefore, prelaminar ischemia alone is unlikely to explain the excavation of a glaucomatous ONH.

Additionally, Hayreh (1974) has argued that primary ischemia to the retrolaminar optic nerve alone could cause optic neuropathy indistinguishable from human glaucoma. However, most investigators believe that ischemic or compressive lesions of the retrolaminar optic nerve seldom lead to true glaucomatous excavation of the ONH surface (Quigley and Anderson, 1977a; Trobe et al., 1980). As noted above, our concept is compatible with the notion that primary retrolaminar ischemia can cause non-IOP-related damage to the axons. Furthermore, in some eyes, retrolaminar ischemia may be capable of altering the material properties of the load-bearing connective tissues such that the onset and progression of mechanical failure occur at normal levels of IOP.

8. ONH biomechanics: experimental studies

8.1. Compliance testing the ONH surface in normal monkeys

A number of investigators have previously shown that acute increases in IOP cause posterior deformation of the surface of the ONH relative to the peripapillary retina (Zeimer and Ogura, 1989; Coleman et al., 1991; Quigley and Pease, 1996; Morgan et al., 2002). We performed initial studies in the laboratories of Dr. Harry Quigley using a video camera to image the monkey ONH surface during acute IOP elevations (Burgoyne et al., 1994, 1995a). Fig. 8A explains the protocol and data for a single compliance test. The position of the ONH surface at each time point of a compliance test is characterized by the image analysis-based parameter *mean position of the disc* (MPD) (Burgoyne et al., 1995a). In these studies, 51 compliance tests were performed on 28 normal eyes of 18 monkeys (Burgoyne et al., 1995a). Fig. 8D presents the mean change from $MPD_{Baseline}$ at each time point of the compliance test for all 51 tests. The results suggest that the normal monkey ONH exhibits a posterior deformation of approximately 30 μm during acute IOP elevations from 10 to 45 mm Hg.

More recently, we examined ONH surface compliance using confocal scanning laser tomographic (CSLT) imaging (Heickell et al., 2001). In these studies, a total of 156 compliance tests were performed on 48 normal eyes in 30 monkeys using three separate protocols. CSLT images were obtained using a 10° and/or 15° and/or 20° scan angle at various times after IOP was raised

from 10 to 30 and/or to 45 mm Hg. At each time point, six images were analyzed to calculate MPD, which expressed the overall position of the optic disc surface at that time point. Statistical analysis (ANOVA) was performed to evaluate differences in the amounts of deformation in individual eyes at different IOPs, at different compliance-testing sessions, and in the two eyes of individual monkeys under the same testing conditions.

In this group of eyes, posterior deformation of the surface of the ONH ranged from 15 to 86 μm as early as 10 min after IOP was increased from 10 to 30 mm Hg (Table 1). When IOP was increased from 30 to 45 mm Hg in a subset of these eyes, most showed additional deformation. Of the 12 eyes for which both 15° and 20° images were obtained during the same compliance test, seven showed larger amounts of deformation in the 20° images (Fig. 9). Of the 18 monkeys tested in both eyes, 12 showed some differences and four showed substantial differences between the two eyes.

From these studies, we concluded that in the normal monkey eye, the surface of the ONH deforms rapidly in response to increased IOP, sometimes in as little as 10 min. The amount of deformation varies from one individual to another and even within the two eyes of individual monkeys. Increasing the scan angle from 15° to 20° frequently increases the amount of deformation detected, suggesting that, in addition to the ONH, the peripapillary sclera may be involved in the deformation in some eyes (Fig. 9).

8.2. The lamina cribrosa and scleral canal deform following acute elevations of IOP

Deformation of the lamina cribrosa following acute IOP elevation has been described in human cadaver eyes (Quigley and Green, 1979; Yan et al., 1994; Albon et al., 2000). To more fully characterize the response of the anterior scleral canal and lamina cribrosa to acute IOP elevation, we performed measurements of anterior laminar position and thickness, scleral canal diameter at Bruch's membrane and the anterior laminar insertion, and anterior scleral canal wall geometry in every sixth 4- μm serial sagittal section of the ONH from both eyes of 17 monkeys (Fig. 10). Under pentobarbital anesthesia, both eyes of four normal monkeys (two normal eyes) and five monkeys with early glaucoma (one normal and one early glaucoma eye) were cannulated, and the IOP set to 10 mm Hg in one normal eye and 30 or 45 mm Hg in the other (normal or early glaucoma) eye. After 15–80 min in which the mean blood pressure (diastolic plus one third the diastolic/systolic difference) was maintained as high as possible (80–100 mm Hg), each monkey was perfusion-fixed with 1 L of paraformaldehyde and 6 L of glutaraldehyde pressurized to 150–200 mm Hg. The ONH tissues from each eye were

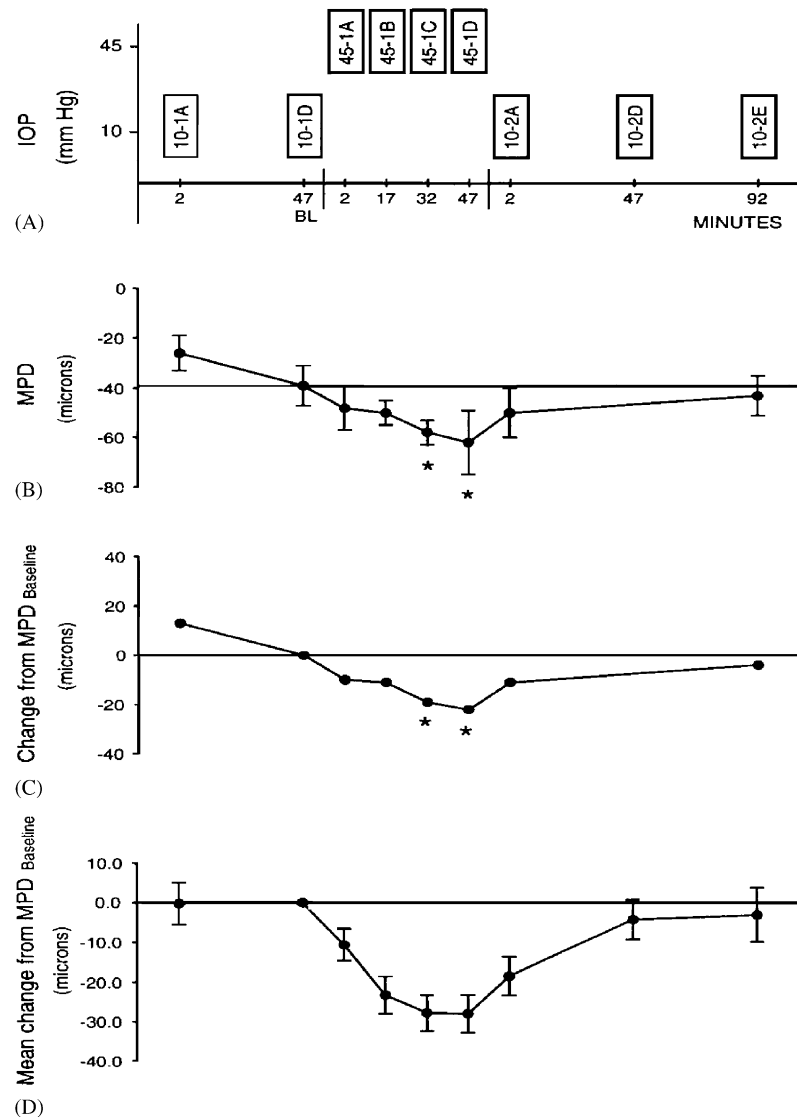


Fig. 8. Compliance testing the normal monkey ONH. (A) Eight images were acquired at each time-point of the test; time-point 10-1D (47 min after IOP was lowered to 10 mm Hg) is the BASELINE time-point to which all others were compared. (B) MPD data from a representative test, presented as Change from MPD_{Baseline} data in (C). (D) Mean Change from MPD_{Baseline} data at each time-point for all 51 compliance tests performed in the study. Adapted from Burgoyne, C.F., Quigley, H.A., Thompson, H.W., Vitale, S., Varma, R., 1995. Measurement of optic disc compliance by digitized image analysis in the normal monkey eye. *Ophthalmology* 102, 1790–1799, Copyright, 1995, with permission from *American Academy of Ophthalmology*.

trephined, embedded, serially sectioned, stained, and digitally imaged. The experiments are outlined in Figs. 11 and 12. The principal findings are as follows.

8.2.1. The lamina cribrosa and scleral canal wall act like an expandable trampoline at low levels of IOP, with the canal expanding and the lamina thinning and more tautly stretched as IOP is elevated from 0 to 10 mm Hg

We first compared pooled data from eight normal monkey eyes immersion-fixed at IOP 0 mm Hg with data from nine normal monkey eyes perfusion-fixed at IOP 10 mm Hg (Table 2). The results demonstrated that the anterior laminar position was significantly more anterior (pulled forward toward Bruch's membrane), the

lamina was thinner, and the scleral canal diameter was larger in the IOP 10 mm Hg eyes, compared with the IOP 0 mm Hg eyes (Fig. 13) (Bellezza et al., 2003a).

8.2.2. The lamina cribrosa bows posteriorly following acute elevation of IOP from 10 to 30 mm Hg

In a follow-up publication (Bellezza et al., 2003b), we performed an experiment-wide ANOVA on the data from all 34 eyes of the 17 monkeys outlined in Figs. 11 and 12. That analysis generated 95% confidence intervals (CIs) for the overall treatment effects of “between-eye physiologic difference,” and “normal laminar compliance,” on each of our measured parameters (Table 3). In this analysis, a small but significant

Table 1
Monkey characteristics and early IOP-30 mm Hg deformation data

Study no.	Monkey no.	Weight (kg)	Axial length (mm) ^a		Disc area (mm ²) ^b		MPD _{Baseline} (μm) ^c		Magnitude of early IOP-30 deformation (μm) using 15° scan angle ^d	
			OD	OS	OD	OS	OD	OS	OD	OS
1	1	5.5	18.64		1.03		+38		−68	
	2	12.0	20.16		1.26		−104		−27	
	3	6.4	21.35		1.88		−156		−34	
2	4	11.4	21.02		1.48		−100		−38	
	5	9.2	20.34		1.53		−115		−42	
	6	7.8	20.19		1.79		−17		−37	
	7	6.5		21.99	1.90			−126		+2
	8	5.5	21.09		1.60		+74		−36	
	9	3.1	19.57		1.31		−154		−9	
	10	12.4	20.38		1.46		+10		−16	
	11	10.9	21.10		1.84		−83		−15	
	12	11.5	20.90		1.71		+26		−86	
3	13	6.7	19.21	19.33	1.49	1.33	−139	−164	−36	−45
	14	6.0	20.87	20.80	1.59	1.76	−127	−132	−10	−8
	15	7.5	21.39	21.25	1.77	1.47	+35	+40	−37	−46
	16	6.8	20.33	20.22	1.93	1.77	−174	−188	−50	−22
	17	6.8	20.75	20.75	1.70	1.80	−151	−146	−35	−33
	18	5.7	20.47	20.47	1.36	1.40	−132	−114	−10	−22
	19	6.0	19.92	19.63	1.54	1.52	−104	−125	−34	−49
	20	6.8	19.74	19.81	1.23	1.23	−86	−102	−64	−54
	21	7.5	21.79	21.54	1.92	1.94	−119	−115	−60	−41
	22	5.8	20.61	20.51	1.62	1.75	−144	−153	−40	−35
	23	6.0	20.40	20.44	1.73	1.43	−165	−132	−9	−51
	24	7.2	20.06	18.71	1.18	1.03	−41	−25	−44	−43
	25	8.3	21.61	21.52	1.65	1.47	−28	−28	−48	−21
	26	8.3	20.72	20.64	1.21	1.26	−121	−93	−39	−41
	27	7.8	21.10	20.84	1.93	1.65	−201	−168	0	−37
	28	5.7	20.30	20.15	1.26	1.27	−123	−99	−42	−42
	29	5.6	21.05	21.00	1.60	1.65	−74	−41	−38	−36
	30	5.0	20.93	20.80	2.27	2.14	−211	−194	+2	−11

Reprinted with permission from Heickell, A.G., Bellezza, A.J., Thompson, H.W., Burgoyne, C.F., 2001. Optic disc surface compliance testing using confocal scanning laser tomography in the normal monkey eye. *J. Glaucoma* 10, 369–382.

^aMean axial length at the baseline observation point for the three or five test days.

^bMean value of the parameter *Disc Area* within the 15° baseline observation point images for the three or five test days.

^cMean value of the parameter *MPD* within the 15° baseline observation point images for the three or five test days.

^dBold type indicates values statistically significantly different from corresponding baseline observation point value ($P < 0.05$, ANOVA).

treatment effect of 12–19 μm for normal laminar compliance exceeded the 95% CI for physiologic between-eye differences. The relatively small effect is likely a result of low IOP elevations (from 10 to 30 mm Hg) of short duration (15 min) in three of the four monkeys.

8.3. Damage to the ONH connective tissues occurs early in experimental glaucoma

8.3.1. Early onset of hypercompliance of the ONH surface in experimental glaucoma

In a second series of experiments performed in Dr. Quigley's laboratories, monkeys were tested for compliance as normals and then on multiple occasions after

the onset of experimental glaucoma or optic nerve transection in one eye (Burgoyne et al., 1995b). In both longitudinally and cross-sectionally studied eyes, ONH surface compliance increased significantly 1–2 weeks after the onset of experimental glaucoma (Fig. 14), and the increase persisted through the 9–12 week evaluation. However, compliance was again reduced during weeks 13–18 and beyond week 18 following IOP elevation.

8.3.2. Relative stiffness rather than hypercompliance in optic nerve transection eyes

In striking contrast to the experimental glaucoma eyes, the optic nerve transection eyes showed no early period of hypercompliance. Instead, the transection eyes

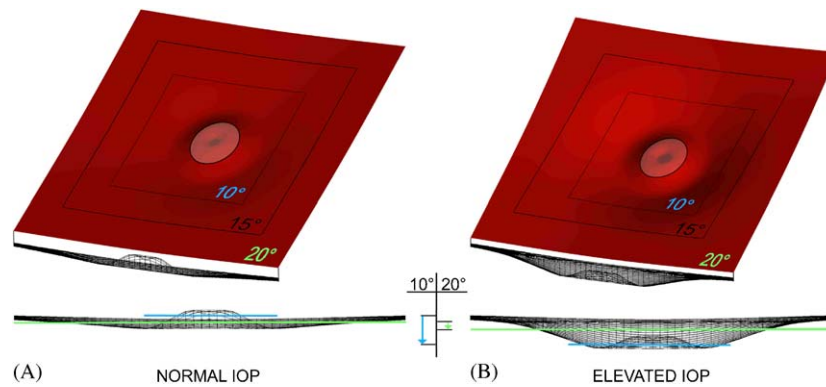


Fig. 9. Posterior deformation of the peripapillary sclera and overlying retina following acute elevation of IOP likely explains the fact that acute deformation within the 20° CSLT images exceeded that seen in the 10° CSLT images. (A) A schematic representation of a 20° CSLT image obtained at an IOP of 10 mm Hg onto which the boundaries of 10° and 15° images have been superimposed. The area of interest (black edge around the pale central region) is the outline of the disc margin, which is the projection of the deeper scleral canal opening (not shown) onto the surface of the peripapillary nerve fiber hump. Below the planar representation, a cross-section of the surface through the middle of the scleral canal shows the 10° (blue) and 20° (green) zero reference planes. Note that the 10° zero reference plane is anterior to the 20° zero reference plane for this eye. (B) Schematic of the same eye at an elevated IOP (either 30 or 45 mm Hg) with posterior deformation of the scleral canal and peripapillary sclera that extends just beyond the boundaries of the 10° image. Note that for a given deformation of the peripapillary sclera and overlying retina, a larger percentage of the 10° zero reference plane pixels and a smaller percentage of 20° zero reference plane pixels are affected. Below the planar representation, a cross-section of the surface of the mid-scleral canal shows the new position of the elevated IOP 10° (blue) and 20° (green) zero reference planes. Note that the resultant posterior shift of the zero reference plane relative to its position in (A) is greater for the 10° images (blue arrow) and less for the 20° images (green arrow). Owing to this differential posterior shift of the zero reference plane, at an elevated IOP the parameter MPD—the average of all elevation values for those pixels contained within the area of interest—will be more negative in the 20° images than in the 10° images relative to its value when IOP is 10 mm Hg. Reprinted with permission from Heickell, A.G., Bellezza, A.J., Thompson, H.W., Burgoyne, C.F., 2001. Optic disc surface compliance testing using confocal scanning laser tomography in the normal monkey eye. *J. Glaucoma* 10, 369–382.

progressed to qualitatively less movement of the disc 5, 9, and 13 weeks following transection (Fig. 14).

8.3.3. Early onset of permanent posterior deformation of the ONH surface in experimental glaucoma

Separate from a change in optic disc compliance, the glaucomatous optic discs that were studied longitudinally underwent an early, chronic deformation of the ONH surface (deformation of the baseline position of the ONH surface between tests) that was not reversed with subsequent lowering of IOP. Optic nerve transection eyes, however, demonstrated no statistically significant change in baseline position of the ONH surface 5, 9, or 13 weeks post-transection (Table 4).

8.3.4. Permanent posterior deformation of the anterior scleral canal wall and lamina cribrosa underlie the onset of CSLT-detected ONH surface change in early experimental glaucoma

This finding is demonstrated in Fig. 12 (above) (Bellezza et al., 2003b). Permanent posterior deformation of the ONH connective tissues is important because it provides the first evidence in support of our central hypothesis that the load-bearing connective tissues of the anterior ONH are damaged early in experimental glaucoma. It has clinical significance in that it suggests that in an ocular hypertensive or glaucoma-suspect eye that is followed until ONH surface change is detected by CSLT imaging, apart from the degree of axonal and

visual field loss, the underlying connective tissues are damaged and the disc should be more susceptible to whatever level of IOP it subsequently experiences.

In a second group of three monkeys with experimental IOP elevations in one eye, permanent posterior deformation of the anterior scleral canal wall and lamina cribrosa were present at the onset of CSLT-detected ONH surface change within 3D reconstructions of the ONH connective tissues of the high IOP eye, compared with the contralateral normal eye (Downs et al., 2003a; Hiron et al., 2003). The method of CSLT detection of the onset of ONH surface change in these monkeys is shown in Fig. 15. The method of 3D reconstruction is explained in Fig. 16. Representative findings in vertical and horizontal central and peripheral digital sections of the normal and early glaucoma eye of one monkey are shown in Fig. 17.

8.3.5. Hypercompliance of the anterior scleral canal wall and lamina cribrosa accompanies the onset of CSLT-detected ONH surface change in early experimental glaucoma

This finding is demonstrated in Fig. 11 above, and is described in detail in a previous report (Bellezza et al., 2003b). It is important because it confirms our previous finding of ONH surface hypercompliance in a separate group of monkeys, and establishes that hypercompliance of the lamina is the underlying cause.

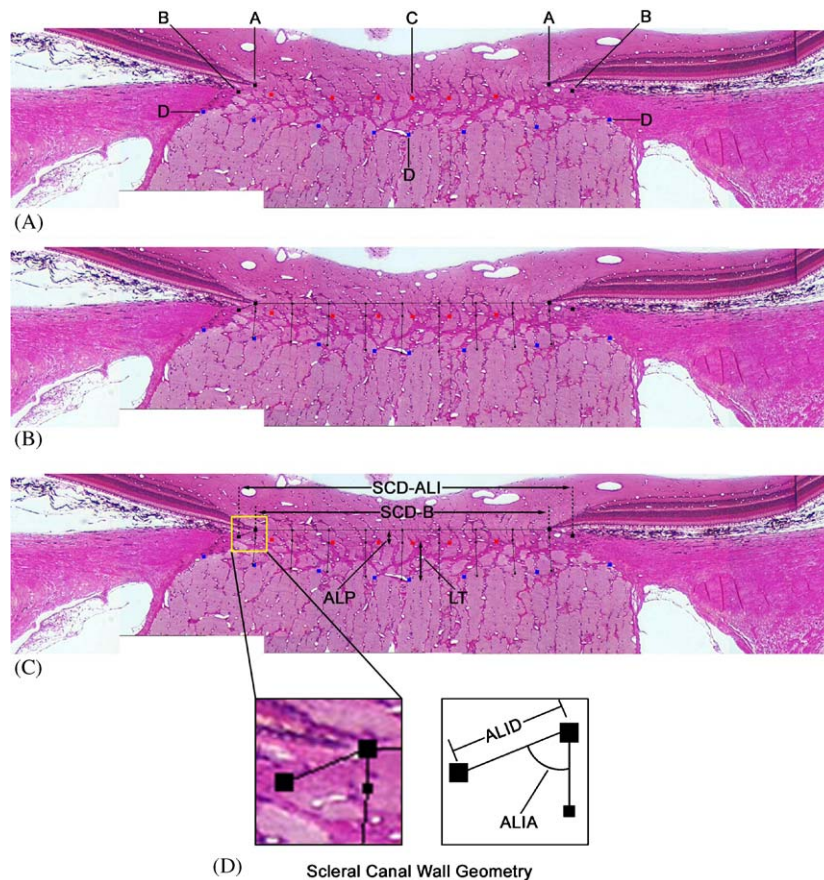


Fig. 10. Measurements within a single sagittal section. From 37 to 67 sections were measured per eye, depending on whether the ONH specimen was sectioned vertically or horizontally. (A) Each composite section image consists of four to six individual overlapping images taken at a resolution of approximately $2\mu\text{m}$ per pixel. On each section image, an operator placed marks denoting the termination of Bruch's membrane (A), the anterior insertion of the lamina into the sclera (B), and the anterior (C) and posterior (D) borders of the lamina cribrosa. (B) Once the section was landmarked, custom image analysis software connected the two Bruch's membrane termination points, divided the distance into nine measurement points, and dropped perpendicular lines to the anterior and posterior lamina cribrosa as best determined by the image analysis software based on the operator's landmarks. (C) Automatic measures of the following parameters were generated: nine measurements of anterior lamina position (ALP) and lamina thickness (LT) across the scleral canal opening and one measurement of the diameter of the scleral canal opening at Bruch's membrane (SCD-B) and at the anterior lamina insertion (SCD-ALI). (D) Additional measures of anterior lamina insertion distance (ALID) and anterior lamina insertion angle (ALIA) were made relative to the scleral canal opening at Bruch's membrane along each side of the canal. Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Deformation of the lamina cribrosa and anterior scleral canal wall in early experimental glaucoma. *Invest. Ophthalmol. Vis. Sci.* 44, 623–637.

Thus, in early glaucoma, the connective tissues of the anterior scleral canal wall and lamina cribrosa demonstrate not only permanent posterior deformation (Fig. 12, above) but also hypercompliance, a separate manifestation of connective tissue damage. The histologic findings and molecular mechanisms that underlie both permanent and hypercompliant connective tissue deformation remain to be characterized.

8.4. The viscoelastic material properties of posterior sclera are altered in early glaucoma monkey eyes

8.4.1. Elastic and viscoelastic material properties of normal rabbit and monkey sclera

Our uniaxial scleral strip testing apparatus and protocol are described in Fig. 18. Data from 64

specimens (eight per peripapillary scleral quadrant) from 64 eyes of 32 rabbits and 24 specimens (six per peripapillary scleral quadrant) from 24 eyes of 16 monkeys have been previously reported (Downs et al., 2001b, 2003b). In these studies, the elastic and viscoelastic material properties of normal monkey peripapillary sclera were not different by quadrant. However, monkey peripapillary sclera was stiffer, relaxed more slowly, and maintained higher equilibrium stresses than rabbit peripapillary sclera.

8.4.2. Viscoelastic material properties of posterior sclera are altered in early glaucoma monkeys eyes

So far, we have accumulated data from eight specimens (four specimens each from the superior and inferior quadrants only) from eight early glaucoma eyes

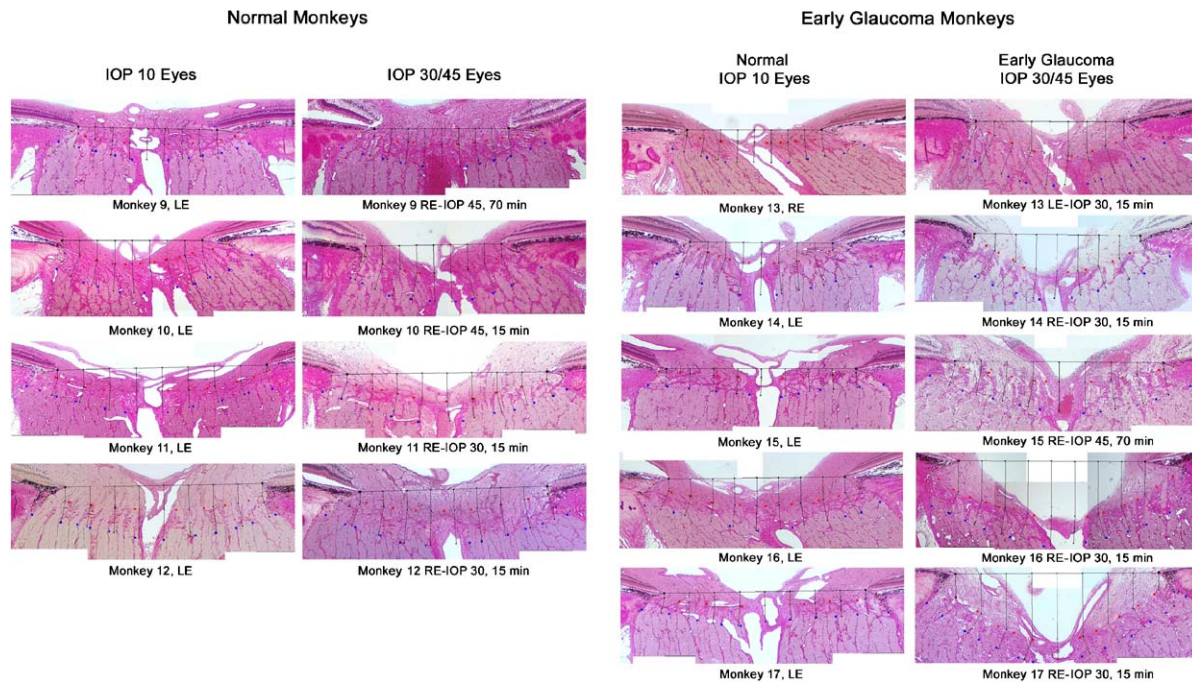


Fig. 11. Central sagittal sections of both eyes of nine perfusion-fixed monkeys (one normal eye at IOP 10 mm Hg, the other normal or early glaucoma eye at IOP 30 or 45 mm Hg) in which measurements of anterior lamina position, lamina thickness, and anterior scleral canal geometry were made. Between-eye differences in the normal monkeys (left) illustrate deformation of the normal lamina following acute IOP elevations from 10 to 30 or 45 mm Hg for 15–80 min. Between-eye differences in the early glaucoma monkeys (right) illustrate both permanent posterior deformation owing to early glaucomatous damage and hypercompliance following similar acute IOP elevations from 10 to 30 mm Hg for 15–70 min. LE, left eye; RE, right eye. Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Deformation of the lamina cribrosa and anterior scleral canal wall in early experimental glaucoma. *Invest. Ophthalmol. Vis. Sci.* 44, 623–637.

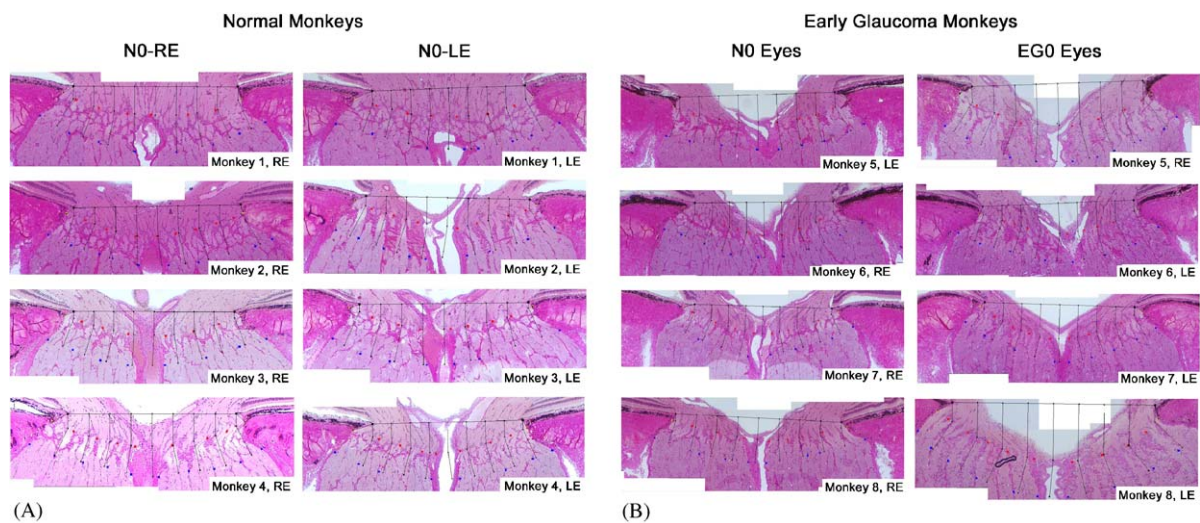


Fig. 12. Central vertical sagittal sections of both eyes of eight monkeys that were enucleated and then immersion fixed (IOP 0 mm Hg). The differences between the two eyes of each normal monkey (A, both eyes normal) characterize the magnitude of physiologic between-eye differences for each of our measurement parameters (treatment differences needed to exceed physiologic between-eye differences in order to be significant). The differences between the two eyes of each early glaucoma monkey (B, normal eye left, early glaucoma eye right) characterize the magnitude of permanent posterior deformation. LE, left eye; RE, right eye. Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Deformation of the lamina cribrosa and anterior scleral canal wall in early experimental glaucoma. *Invest. Ophthalmol. Vis. Sci.* 44, 623–637.

of eight monkeys (Downs et al., 2001b). These data suggest that the long-term viscoelastic (but not instantaneous elastic) material properties of monkey peripa-

pillary sclera are altered within 3–6 weeks of chronic exposure to moderately elevated IOP. Peripapillary sclera from early glaucoma eyes relaxed more slowly

Table 2

Lamina cribrosa position and thickness by IOP group^a

Region	Anterior laminar position		Laminar thickness	
	IOP 10 mm Hg	IOP 0 mm Hg	IOP 10 mm Hg	IOP 0 mm Hg
Overall	116 ± 2	184 ± 2	195 ± 2	264 ± 2
Central	161 ± 4	267 ± 6	223 ± 4	268 ± 4
Superior	102 ± 2	151 ± 4	179 ± 2	273 ± 4
Inferior	106 ± 4	158 ± 4	186 ± 2	270 ± 2
Nasal	109 ± 4	181 ± 4	189 ± 2	258 ± 4
Temporal	102 ± 2	162 ± 4	199 ± 2	252 ± 2

Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Anterior scleral canal geometry in pressurised (IOP 10) and nonpressurized (IOP 0) normal monkey eyes. Br. J. Ophthalmol. 87, 1284–1290 (<http://bjo.bmjjournals.com/>).

^aValues are means ± 95% confidence intervals in μm .

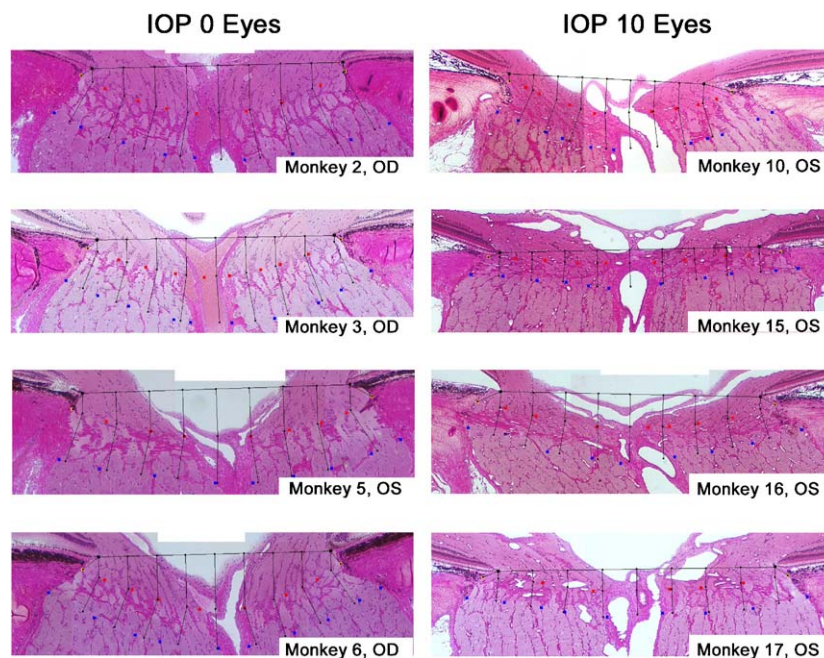


Fig. 13. Representative middle sagittal section images from four representative IOP 0 mm Hg (immersion-fixed) and four representative IOP 10 mm Hg (perfusion-fixed) ONHs from normal monkey eyes. Note that within the IOP 0 eyes (left), the lamina cribrosa appears “sagged” posteriorly relative to the “taut,” more anteriorly positioned lamina in the IOP 10 eyes (right). OD, right eye; OS, left eye. Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Anterior scleral canal geometry in pressurised (IOP 10) and non-pressurised (IOP 0) normal monkey eyes. Br. J. Ophthalmol. 87, 1284–1290 (<http://bjo.bmjjournals.com/>).

and maintained higher equilibrium stresses than that of normal eyes.

8.5. IOP-related stress and strain markedly diminish patency of the anterior laminar capillaries in normal and early glaucoma monkey eyes at high IOP

To test the hypothesis that moderate, short-term IOP elevations diminish the patency of capillaries in the lamina cribrosa, we qualitatively evaluated anterior and posterior laminar and retrolaminar capillary patency in the high-resolution digital images of peripheral and

central sections from both the low and high IOP eye of each of the nine perfusion-fixed monkeys described in Fig. 11, above.

At the time of enucleation, all nine of the high IOP eyes, but none of the low IOP eyes, contained small amounts of residual blood within the retinal and vortex vessels. Within the digital section images (Fig. 19), the posterior laminar and retrolaminar capillaries were widely patent in all low and high IOP eyes. However, anterior laminar capillary patency was markedly diminished in all nine high IOP eyes, compared to the contralateral low IOP eyes. We concluded that anterior

Table 3
Magnitude of experiment-wide effects

Parameter (μm)	Immersion-fixed monkey eyes		Perfusion-fixed monkey eyes		Hypercompliance alone Difference between [EG30/45–N10(EG)] and [EG0–N0]
	Intramonkey physiologic between-eye difference N0- RE–N0-LE	Plastic deformation EG0–N0	Normal laminar compliance N30/ 45–N10(N)	Permanent deformation with early glaucomatous hypercompliance EG30/45–N10(EG)	
Anterior laminar position	1–17^a	62–77	10–23	132–145	55–83
Laminar thickness	6–15	50–60	3–11	56–64	–4–14
<i>Vertical scleral canal diameter</i>					
Bruch's membrane	–51–52	33–129	–69–65	88–198	–41–69
Anterior laminar insertion	–109–7	31–139	–96–56	135–259	–4–228
<i>Horizontal scleral canal diameter</i>					
Bruch's membrane	NA	NA	–46–40	96–182	NA
Anterior laminar insertion	NA	NA	–61–12	23–96	NA

Pairs of values in bold print denote significant differences between the compared groups. N0-RE and N0-LE, normal right and left eyes of normal monkeys, immersion-fixed at IOP 0 mm Hg; N0 and EG0, normal and early glaucoma eyes of early glaucoma monkeys, immersion-fixed at IOP 0 mm Hg; N10(N), normal eye of normal monkey, perfusion-fixed at IOP 10 mm Hg; N30/45, normal eye of normal monkey, perfusion-fixed at IOP 30 or 45 mm Hg; N10(EG), normal eye of early glaucoma monkey, perfusion-fixed at IOP 10 mm Hg; EG 30/45, early glaucoma eye of early glaucoma monkey, perfusion-fixed at IOP 30 or 45 mm Hg; NA, not available. Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Deformation of the lamina cribrosa and anterior scleral canal wall in early experimental glaucoma. Invest. Ophthalmol. Vis. Sci. 44, 623–637.

^aValues are 95% confidence intervals in μm of the differences between the compared groups of eyes.

laminar capillary patency was markedly diminished in normal and early glaucoma monkey eyes perfusion-fixed following moderate, short-term IOP elevations.

Previous studies using microspheres (Alm, 1973; Geijer and Bill, 1979) have suggested that volume flow within the prelaminar and anterior laminar capillary beds is preferentially diminished once perfusion pressure (defined as the mean blood pressure minus the IOP) is less than 30 mm Hg. Although the clinical significance of our findings remains to be determined, they are in agreement with these previous reports and suggest that within the tissues of the normal and early glaucoma monkey ONH, blood flow within the anterior laminar capillaries may be most susceptible to fluctuations in ocular perfusion pressure.

9. Finite element modeling of the ONH

We are building continuum and micro finite element models of the load-bearing connective tissues of the ONH so as to model and predict the stresses and strains they experience. The finite element method is an engineering tool used to study structures with complex geometries and material properties (Fig. 20). In the finite element method, the digital geometry of the structure being modeled is “discretized” into multiple, small, regularly shaped building blocks (the finite elements—

see Fig. 20). The mechanical behavior of the total structure is then calculated from the combined behavior of these constitutive elements. As applied to the lamina cribrosa and scleral canal, our finite element models will consist of a computational model that captures the important aspects of their geometry, material properties, and boundary conditions.

Because each element of the model is individually treated (i.e., programmed to possess its own shape and material properties), computing requirements dictate that structures be modeled first using a “continuum” approach, in which the digital geometry of the lamina cribrosa, for example, is broken into relatively large representative volume elements that include multiple laminar beams (Fig. 21). For each continuum element, the stiffness is found from the volume fraction and predominant orientation of the beams contained within the element. Model development advances from continuum to micro finite element modeling by gradually discretizing the large continuum elements into their constituent voxels or groups of voxels (Fig. 21). In related voxel-based finite element models of trabecular bone, the individual voxels within each trabecular strut themselves become elements, thus generating models with 10–100 million elements, requiring 20,000 CPU hours (6 weeks) of supercomputer time to characterize the stresses and strains within each element of the model (Quigley et al., 1982a).

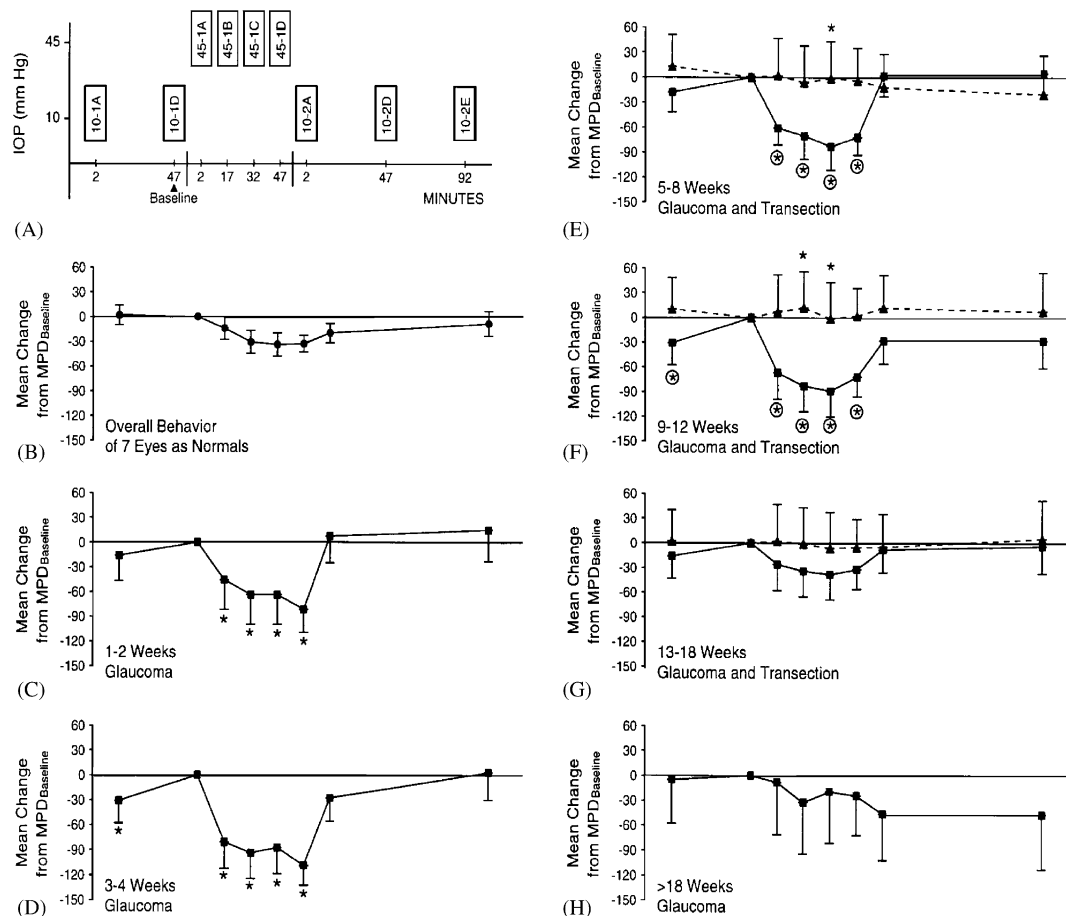


Fig. 14. Onset of ONH hypercompliance in glaucoma but not optic nerve transection eyes. (A) Schematic showing time-points of the testing. (B) Mean change from MPD_{Baseline} behavior of study eyes as normals ($-30\mu\text{m}$ of movement). (C, D) Onset and progression of hypercompliance; -90 , then $-120\mu\text{m}$ of posterior movement as early as 1–2 weeks after the onset of elevated IOP. (E–G) Optic nerve transection eyes (dotted lines) not only show no hypercompliance but are rigid; hypercompliance in glaucoma eyes has evolved to relative rigidity by week 13–18. (H) Relative rigidity beyond week 18. Time points marked with asterisk display change significantly different ($P < 0.05$, ANOVA) from movement at this time-point when the study eyes were normal. Time-points marked with a circled asterisk display change significantly different ($P < 0.05$, ANOVA) from movement of optic nerve transection ONHs at this testing interval. Reprinted from Burgoyne, C.F., Quigley, H.A., Thompson, H.W., Vitale, S., Varma, R., 1995. Early changes in optic disc compliance and surface position in experimental glaucoma. *Ophthalmology* 102, 1800–1809, Copyright, 1995, with permission from *American Academy of Ophthalmology*.

Table 4

Magnitude and statistical significance of chronic posterior deformation of the ONH surface in glaucoma and optic nerve transected eyes

Monkey eyes studied as normals	Change from normal MPD _{Baseline} (μm) at given post-intervention testing interval					
	1–2 weeks	3–4 weeks	5–8 weeks	9–12 weeks	13–18 weeks	> 18 weeks
<i>Glaucoma eyes</i>						
Monkey No. 13		+9	–10		–21	
Monkey No. 14	–42	–195*	–126*	–81*	–182*	
Monkey No. 15	–87*	–62*	–133*	–162*	–153*	–87*
Monkey No. 16	–19	–108*	–183*	–212*	–193*	
Monkey No. 17			–110*	–172*		
<i>Transection eyes</i>						
Monkey No. 18			–15	–41	–2	
Monkey No. 19			–37	–28	–27	

_Testing interval during which study eye was compliance tested. * $P < 0.05$, ANOVA. Reprinted from Burgoyne, C.F., Quigley, H.A., Thompson, H.W., Vitale, S., Varma, R. 1995. Early changes in optic disc compliance and surface position in experimental glaucoma. *Ophthalmology*, 102, 1800–1809, Copyright, 1995, with permission from *American Academy of Ophthalmology*.

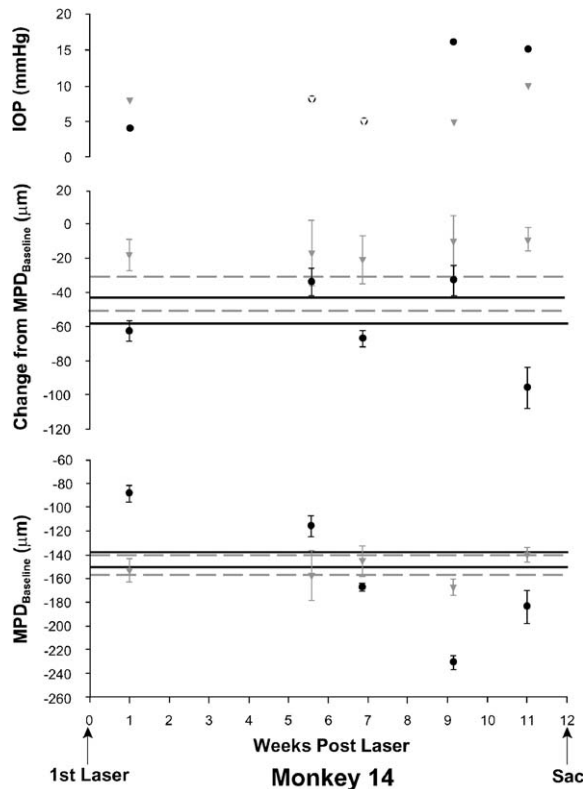


Fig. 15. Post-laser IOP, Change from $MPD_{Baseline}$, and $MPD_{Baseline}$ data from both eyes of a representative early glaucoma monkey. IOP, Change from $MPD_{Baseline}$ (surface compliance following an acute IOP elevation from 10 to 30 mm Hg), and $MPD_{Baseline} \pm 95\%$ confidence interval (surface deformation) are plotted for both the normal (triangles) and early glaucoma eyes (circles) at each post-laser imaging session. For all three parameters, the horizontal lines represent the upper and lower boundaries of the 95% CI for the mean of the three pre-laser, normal compliance-testing sessions (dashed lines, normal eye; solid lines, early glaucoma eye). This monkey was killed (Sac) at the onset of confocal scanning laser tomography-detected ONH surface change, which was defined as the onset of permanent posterior deformation of the ONH surface ($MPD_{Baseline} \pm 95\%$ CI beneath the normal pre-laser range) at two consecutive post-laser imaging sessions. However, in this monkey, because the $MPD_{Baseline}$ data at week 7 post-laser were only questionably significant, the experimental glaucoma eye was followed for one additional imaging session. Reprinted with permission from Bellezza, A.J., Rintalan, C.J., Thompson, H.W., Downs, J.C., Hart, R.T., Burgoyne, C.F., 2003. Deformation of the lamina cribrosa and anterior scleral canal wall in early experimental glaucoma. *Invest. Ophthalmol. Vis. Sci.* 44, 623–637.

The principal goal of building finite element models is to acquire the ability to estimate the mechanical response of the ONH connective tissue structure, i.e., its deformed shape, as well as its constitutive stresses and strains, under a variety of loading conditions. In the case of the normal and early glaucoma ONH, we will refine and validate our models' predictions of lamina deformation using the rigorous histologic studies outlined previously (Bellezza et al., 2003a,b), and the deformation within the 3D reconstructions of a new

group of normal and early glaucoma monkeys that have been perfusion-fixed at varying levels of IOP (Table 5). Demonstration of the accuracy of the models' predictions—a key component of the refinement and validation process—will serve as evidence that engineering principles can be used to understand and predict ONH connective tissue behavior in normal, early, and moderate glaucoma monkey eyes.

9.1. Idealized models

Our initial modeling used finite element models from 13 digital 3D geometries, representing the posterior scleral shell of 13 idealized human eyes (Fig. 22) (Bellezza et al., 2000). The purpose of this study was to assess the effect of the size and shape of the scleral canal, as well as the thickness of the peripapillary sclera, on the level of IOP-related stress for a given level of IOP. Within these models, larger scleral canal diameter, elongation of the canal, and thinning of the sclera all increased IOP-related stress for a given level of IOP. For all models, maximum IOP-related stress ranged from 6 times IOP (posterior sclera) to 122 times IOP (lamina trabeculae). For each model, maximum IOP-related stress was highest within the lamina trabecular region, and decreased progressively through the lamina insertion, peripapillary scleral, and posterior scleral regions. Varying the inner radius had little effect on the maximum IOP-related stress within the scleral canal.

These initial finite element models suggest that IOP-related stress within the load-bearing connective tissues of the ONH is substantial even at low levels of IOP, which supports our concept that the ONH is a “high-stress” environment that requires strong connective tissues to withstand the stresses generated by even normal levels of IOP. These data additionally suggest that scleral canal size/shape and scleral thickness are principal determinants of the magnitude of IOP-related stress within the connective tissues of the ONH for a given level of IOP. However, models that incorporate physiologic scleral canal and lamina geometries, a more refined finite element model meshwork, and non-isotropic material properties are needed to improve clinical relevance.

9.2. Continuum and micro finite element models

Continuum and micro finite element modeling of the ONH requires construction of 3D digital geometries that more accurately represent the 3D architecture of the structure being modeled. We are currently constructing representative geometries for the posterior scleral shell, peripapillary retina, scleral canal, and lamina cribrosa that are based on the actual anatomy of individual monkey eyes.

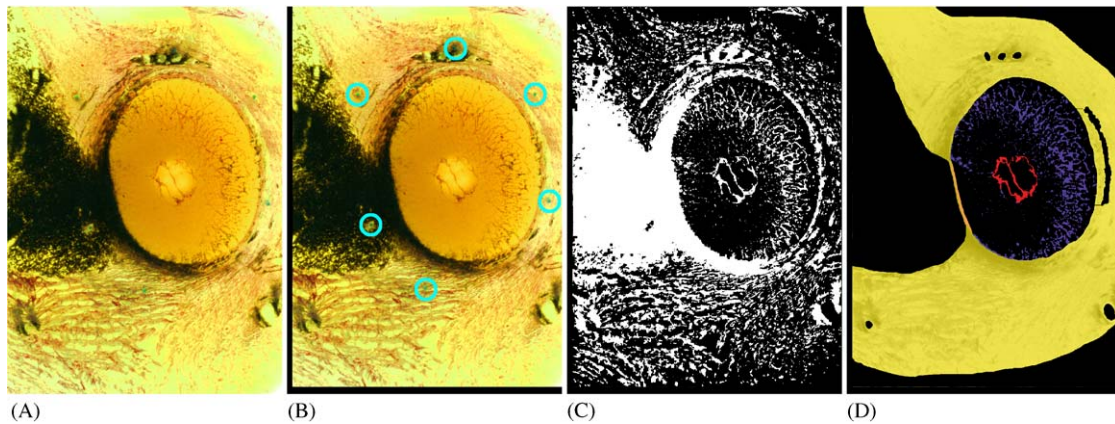


Fig. 16. Stained-section aligned, thresholded, and color-mapped object-delineated images. To acquire an individual stained-section image (A), the camera is positioned relative to the microtome so that the image of the stained, centered, embedded tissue block surface is acquired at a resolution of $2.5 \times 2.5 \mu\text{m}$ per pixel after each $3 \mu\text{m}$ section is cut away. Custom software is then used for image inspection, naming, and storage. Stain is again applied to the embedded tissue block surface and the next stained-section image is acquired. (B) Aligned stained-section image—note the preplaced sutures (blue-circled), which are used to align all of the serial section images. (C) Aligned image thresholded to isolate the connective tissues. (D) Color-mapped object-delineated image in which automated thresholding and edge detection are used by the operator to isolate and identify the following structures (if present) within each image: sclera (yellow), lamina cribrosa (periwinkle); pial sheath (light blue); central retinal vessels (red). Eventually, additional structures such as the posterior ciliary arteries and the laminar capillaries will be delineated as well.

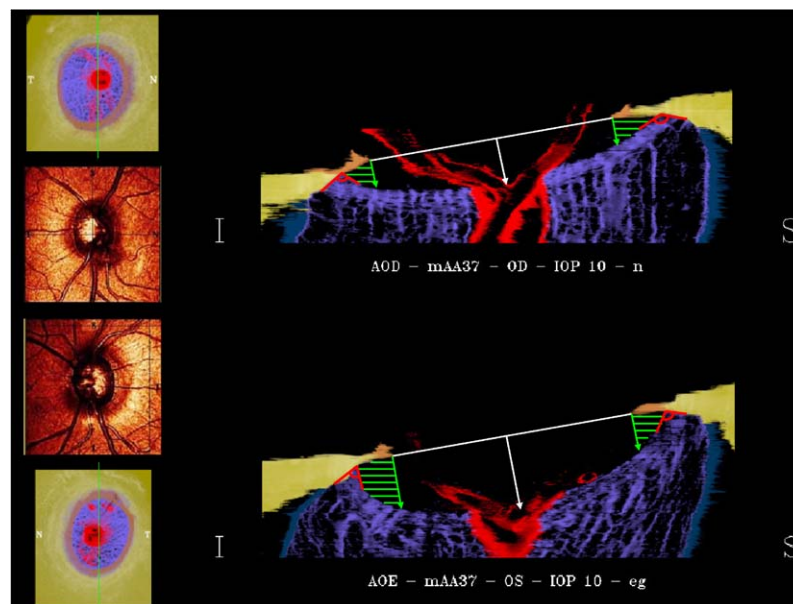


Fig. 17. Central vertical digital sections of the 3D digital reconstructions of the normal (upper panel) and early glaucomatous (lower panel) eyes of a representative early glaucoma monkey. The surface of the 3D reconstruction with a green line marking the location of the digital section and a confocal scanning laser tomographic image of each eye are shown on the left side of each panel. Within each panel, the vertical digital section (inferior, left; superior, right) is displayed on the right. Note the increased central (white arrows) and peripheral (green arrows) lamellar deformation in the early glaucoma eye. Note as well the onset of early excavation (inferior greater than superior) as an increase in the green shaded area beneath the opening in Bruch's membrane in the early glaucoma eye.

9.2.1. Posterior scleral shell surface

The posterior scleral surfaces of normal and early glaucoma monkey eyes perfusion-fixed at IOP 10 mm Hg have been scanned to obtain pooled digital versions of the normal and early glaucoma monkey posterior scleral surface (Fig. 23).

9.2.2. Peripapillary scleral thickness

We have reported peripapillary (0–1500 μm from the scleral canal wall) scleral thickness measurements for nine normal monkey eyes perfusion-fixed at IOP 10 mm Hg (Downs et al., 2002). In all quadrants, the peripapillary sclera was thinnest at the posterior scleral

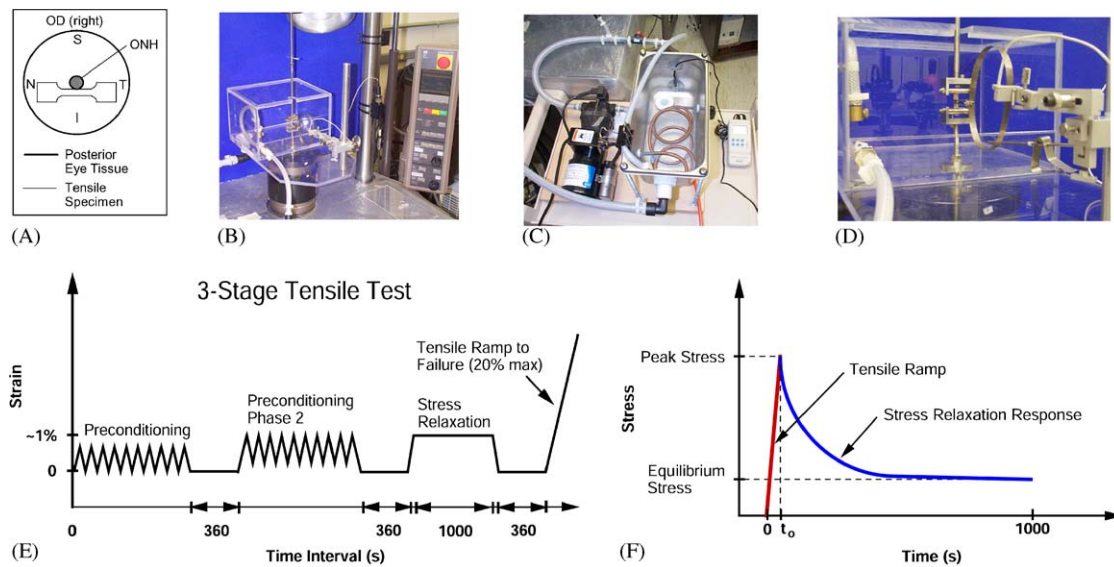


Fig. 18. Uniaxial elastic and viscoelastic strength of materials testing. (A) A specimen cutter was constructed to obtain a single, uniform, dumbbell-shaped specimen from each eye for testing. (B–D) Testing was performed within a custom testing chamber that maintained the specimen in a 37°C, 100% humidity environment throughout the test. (D) An extensometer was employed to measure strain (local deformation) directly from the central portion of the specimen rather than using grip-to-grip displacement to calculate strain indirectly. (E) Time plot of a three-stage tensile test in which the preconditioning, stress-relaxation, and tensile ramp to failure stages are depicted. (F) Standard stress relaxation response of biologic tissue (stage two) from which viscoelastic material property parameters are calculated: instantaneous modulus (measure of tissue's instantaneous resistance to deformation or stiffness), relaxation time constants (how quickly the stress relaxes in response to constant deformation), and equilibrium modulus (measure of long term tissue stiffness after the material has fully stress relaxed). (A), (D), and (E) Reprinted with permission from Downs, J.C., Suh, J.-K.F., Thomas, K.A., Bellezza, A.J., Burgoyne, C.F., Hart, R.T., 2003. Viscoelastic characterization of peripapillary sclera: material properties by quadrant in rabbit and monkey eyes. *J. Biomech. Eng.* 125, 124–131, Copyright, 2003, American Society of Mechanical Engineers (ASME).

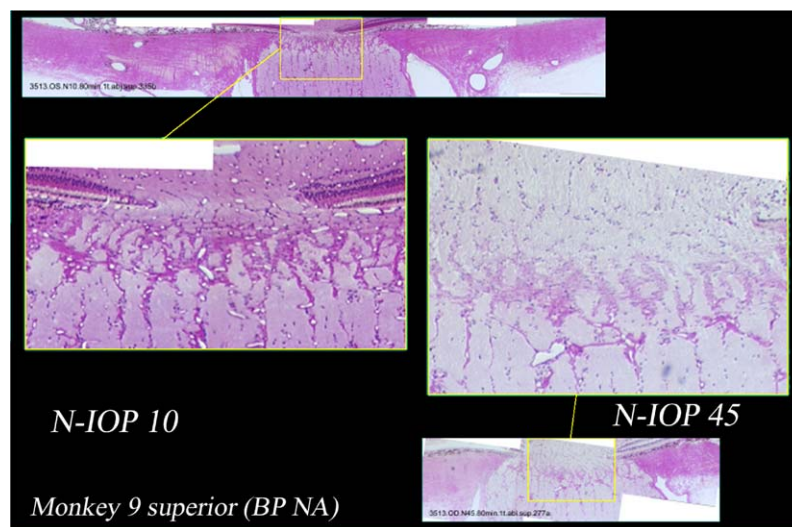


Fig. 19. Histologic sections from the superior scleral canal of both eyes of a normal monkey perfusion fixed with one eye at IOP 10 mm Hg (middle left and above) and one eye at IOP 45 mm Hg (middle right and below). Higher magnification demonstrates patent prelaminar, laminar, and retrolaminar capillaries in the IOP 10 eye (middle left). However, in the IOP 45 eye (middle right), the prelaminar and anterior laminar capillaries do not appear patent, with only the posterior laminar and retrolaminar capillaries open.

canal opening (201 μm —nasal; 201 μm —temporal; 240 μm —inferior; and 249 μm —superior) and progressively thickened to a maximum (326 μm —nasal; 415 μm —superior; 420 μm —temporal; and 422 μm —inferior) 600–1000 μm from the posterior scleral canal opening (Fig. 24).

9.2.3. Posterior scleral thickness

We have reported posterior scleral thickness data from both eyes of eight perfusion-fixed monkeys (four normal IOP 10/30 or 45 mm Hg monkeys and four early glaucoma IOP 10/30 or 45 mm Hg monkeys) (Downs et al., 2001a). In the perfusion-fixed normal monkey

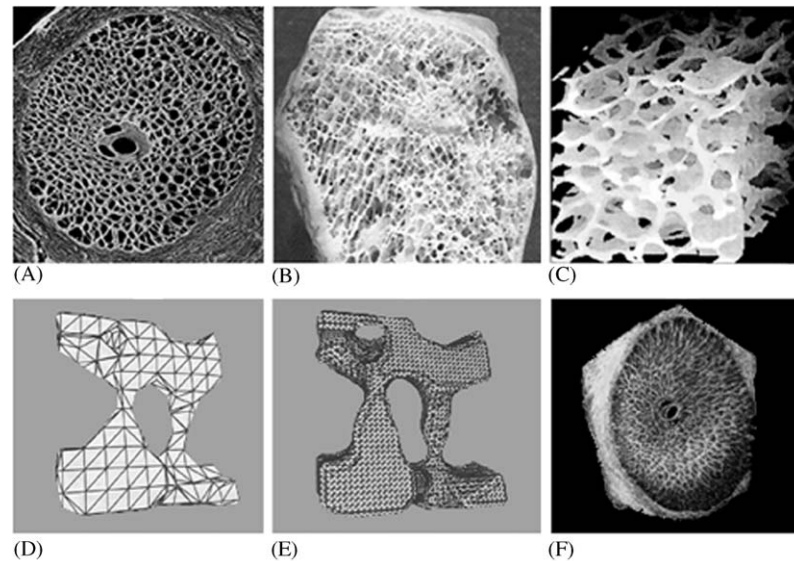


Fig. 20. Finite element modeling. The 3D microarchitecture of the lamina cribrosa (A) is similar to that of trabecular bone (B), which has been modeled using finite element methodology and serves as an example of the process. (C) Finite element modeling starts with a digital 3D geometry that approximates the microarchitecture of the object being modeled, seen here in this digital reconstruction of serial histologic sections of trabecular bone. (D) The 3D geometry of the model is then subdivided into a series of individual finite elements, seen here in a close-up of two individual trabeculae subdivided into their constituent finite elements. (E) Progressive refinement of the model with larger numbers of elements increases its accuracy. (F) An early digital, 3D reconstruction of 64 serial, 5- μ m histologic sections of a monkey optic nerve head, in which the connective tissues of each section have been stained and then isolated by thresholding. (A) Reprinted with permission from Minckler, D.S., 1989. Histology of optic nerve damage in ocular hypertension and early glaucoma (summary). *Surv. Ophthalmol.* 33, 401–402. (B) Reprinted with permission from Bellezza, A.J., Hart, R.T., Burgoyne, C.F., 2000. The optic nerve head as a biomechanical structure: initial finite element modeling. *Invest. Ophthalmol. Vis. Sci.* 41, 2991–3000. (C) Reprinted with permission from van Reithbergen, B., Weinans, H., Huiskes, R., Odgaard, A., 1995. A new method to determine trabecular bone elastic properties and loading using micromechanical finite-element models. *J. Biomechanics* 28, 69–81. (D and E) Reprinted with permission from Müller, R., Hildebrand, T., Rueggsegger, P., 1994. Non-invasive bone biopsy: a new method to analyze and display the three-dimensional structure of trabecular bone. *Phys. Med. Biol.* 39, 145–164. (F) Reprinted with permission from Bellezza, A.J., Hart, R.T., Burgoyne C.F., 2000. The optic nerve head as a biomechanical structure: initial finite element modeling. *Invest. Ophthalmol. Vis. Sci.* 41, 2991–3000.

eyes, the posterior sclera thinned progressively from the fovea to the equator, and was thinnest just posterior to the equator. In the majority of the monkey eyes subjected to acute or short-term chronic IOP elevations, regional thinning was seen in the posterior sclera (Fig. 25).

9.2.4. Construction of a composite posterior scleral shell finite element model

The composite posterior scleral shell curvature has been loaded into Patran finite element model software (MSC Software, Santa Ana, CA) and surfaced. The pooled posterior and peripapillary scleral thickness data are now being added to the inner aspect of the surface and discretized into individual finite elements. A total of three elements will span the full thickness of the wall (Fig. 26).

9.2.5. Construction of a continuum ONH finite element model for each connected, surfaced, 3D geometry

Continuum ONH finite element models for the digital 3D geometries outlined in Table 5 are under construc-

tion. Fig. 21 explains the construction of continuum and micro finite element models. Fig. 27 explains the process of continuum modeling of the normal and early glaucoma eye of one monkey. Fig. 28 demonstrates the output stress and strain distributions within each eye following an acute IOP elevation from 10 to 30 mm Hg. We hypothesize that what makes a given optic neuropathy glaucomatous is that neural and axonal damage are governed by the distribution of IOP-related stress and strain, regardless of the mechanism of damage or the level of IOP at which it occurs (Fig. 29).

9.2.6. Micro finite element modeling

In continuum models, a single element encompasses many laminar beams. In contrast, micro finite element models map stress and strain onto individual laminar beams. Fig. 30 explains our strategy of micro finite element modeling, in which micro finite element models of the laminar beams within individual or adjacent continuum finite elements will be run using the deformations of the parent continuum element(s) faces as the boundary conditions of the contained laminar

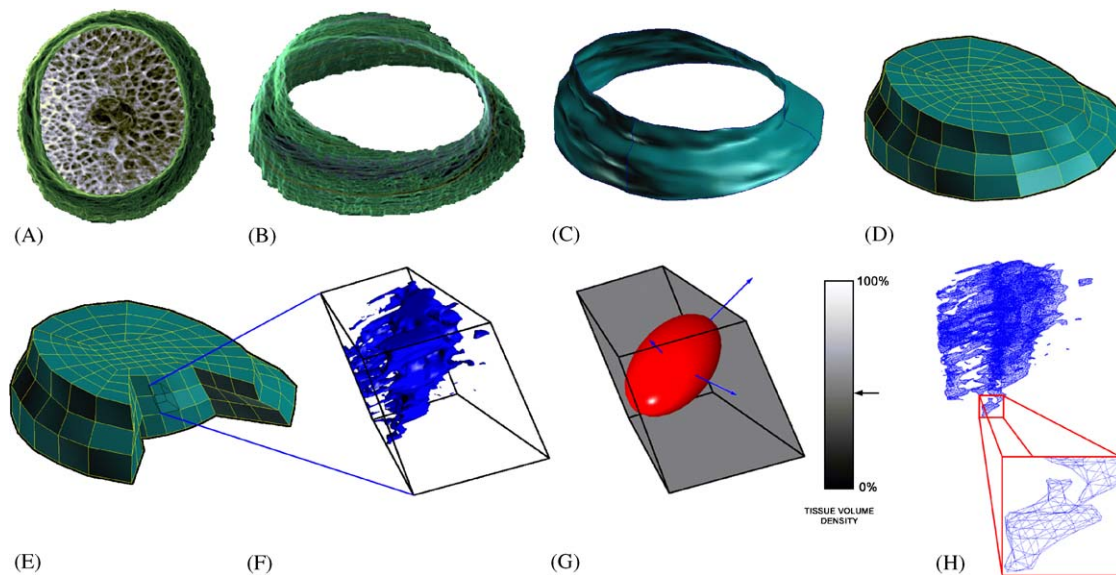


Fig. 21. Continuum and micro finite element model (FEM) construction. (A) The delineated scleral canal wall and lamina are isolated from the voxel geometry of a normal eye. The scleral canal wall alone is isolated in (B), and surfaced in (C) using Patran FEM pre-processing software. Then, the volume within this surface (the intracanalicular space) is filled with 315 finite elements (D). For each constituent element (E and F), material properties are assigned based on the density and predominant orientation of the laminar beams within that element (G). To accomplish this, custom software queries the 3D voxel geometry for information on the presence or absence of connective tissue voxels within the volume of each element (F and G) and casts rays through the element volume to determine the predominant orientation of the laminar beams therein. The material properties of each continuum finite element are a function of the connective tissue volume fraction and the predominant orientation of the connective tissue beams. For the continuum models of the normal and early glaucoma eyes, we will use the uniaxial viscoelastic material properties for posterior monkey sclera that we have previously characterized. In contrast to the large elements of the continuum models, our micro FEMs will use individual voxels or groups of voxels as elements (H).

Table 5
Scheduled 3D reconstructions

Monkey status	Number of monkeys at each IOP (mm Hg)		
	10/10 ^a	10/30	10/45
Normal	4	3	3
Early glaucoma	3	3	3
Moderate glaucoma	4	4	4

^aIOPs at which the two eyes of a given monkey will be set prior to perfusion fixation.

beams. In this manner, micro finite element models of areas of interest within the ONH, as suggested by the continuum modeling of the same ONH, will be run, making both the computational processing and analysis of results more efficient.

Initial results from a micro finite element model yielded average von Mises stresses that were many times those seen in the parent continuum element, with large stress and strain concentrations and gradients seen within individual laminar beams (Fig. 31). Micro finite element modeling of the microarchitecture of the connective tissue of the lamina cribrosa will be used to predict the mechanical environment of the connective tissue beams, containing astrocytes, and adjacent

axons at physiologic and pathophysiologic levels of IOP.

10. Clinical and research implications of the biomechanical paradigm

10.1. Implication 1

For an individual eye, there are at least three components of susceptibility to a given level of 24-h IOP exposure. The first component (Fig. 2) is determined by the relationship between the magnitude of the IOP and the magnitude of the IOP-related stress within the load-bearing ONH connective tissues of that particular eye (Bellezza et al., 2000). The second component (Fig. 3) evolves from the larger issue of how a given ONH tolerates a given level of IOP-related stress. This second component of susceptibility is principally determined by the relationship between the level of IOP-related stress and (1) the laminar and prelaminar axonal nutrient supply and (2) the material properties of the individual lamina cribrosa and peripapillary sclera. The third component of susceptibility (not shown) evolves from the retinal ganglion cell and ONH astrocyte and glial cell response to pathophysiologic stress within the axon (by whatever

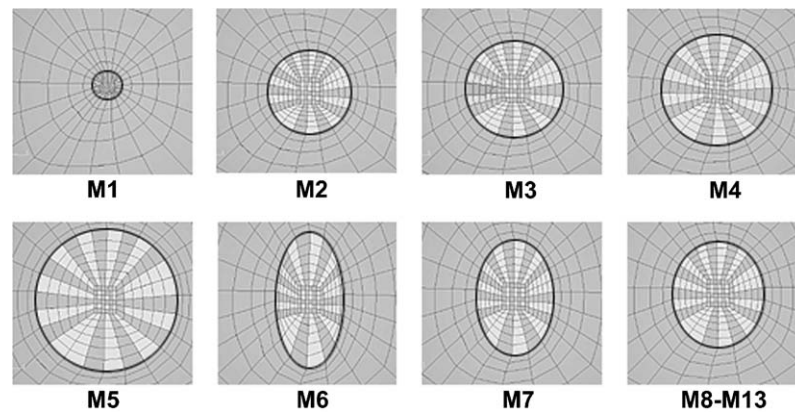


Fig. 22. Dimensions of idealized scleral canal geometries (depicted to scale). A heavy line outlines the scleral canal for all models. Darker shaded elements are those with 'scleral' material properties, whereas lighter shaded elements have 'axonal' material properties. The peripapillary and posterior scleral shell (outside of the scleral canal) are two finite elements thick. Although it is not clearly illustrated in this figure, within the scleral canal there is only a single, posterior (outer) layer of elements depicting the relatively posterior position of the lamina cribrosa as it is situated within the human scleral canal. Note that due to the relative crudeness of the laminar geometry, horizontally aligned laminar beams are consistently thinner than the vertically aligned beams within models containing elliptical scleral canals (M6, M7, and M8–M13). In models 1–8, axial length and peripapillary scleral thickness were kept constant, while the size and shape of the scleral canal were altered. In models 9–13, the size and shape of the scleral canal opening were kept constant while axial length (models M9–M11) or peripapillary scleral thickness (models M12 and M13) was altered. Reprinted with permission from Bellezza, A.J., Hart, R.T., Burgoyne, C.F., 2000. The optic nerve head as a biomechanical structure: initial finite element modeling. *Invest. Ophthalmol. Vis. Sci.* 41, 2991–3000.

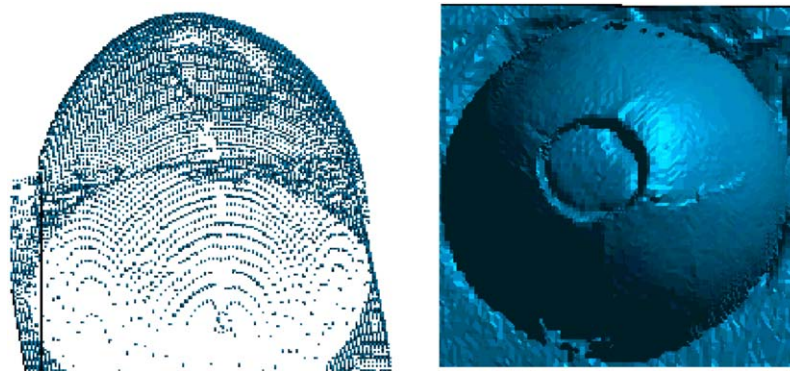


Fig. 23. Digital scans of the posterior surface of a single scleral shell. Left, raw data. Right, smoothed data.

mechanism). Our paradigm suggests that mechanical failure of the connective tissues underlies (and defines) glaucomatous cupping. However, it also acknowledges that the cellular component of susceptibility makes an important contribution to certain aspects of that failure, and likely mediates the vision-threatening insult to the axon in this disease.

Thus, our paradigm suggests that the overall susceptibility of the load-bearing connective tissues to IOP-related stress is determined by (1) the level of IOP-related stress and (2) the material properties of the laminar and scleral canal wall extracellular matrix. It further suggests that the overall susceptibility of the axons to IOP-related stress within the lamina cribrosa is determined by the interplay between (1) the level of IOP-related tensile strain, (2) the laminar capillary perfusion

pressure (i.e., volume flow), (3) the ease of diffusion from the capillary to the axon bundles, and (4) the material properties and diffusion characteristics of the laminar extracellular matrix.

10.2. Implication 2

The clinical appearance of the ONH in glaucoma (pallor vs. glaucomatous excavation) reflects the relative involvement of the neural and/or connective tissues and the physiologic age of the ONH, not the mechanism of insult (Fig. 4). Previous investigators have suggested that the tissues of the ONH can assume several different appearances in glaucoma (Levene, 1980; Caprioli and Spaeth, 1984, 1985; Nicoleta and Drance, 1996). Some have argued that vascular mechanisms predominate in

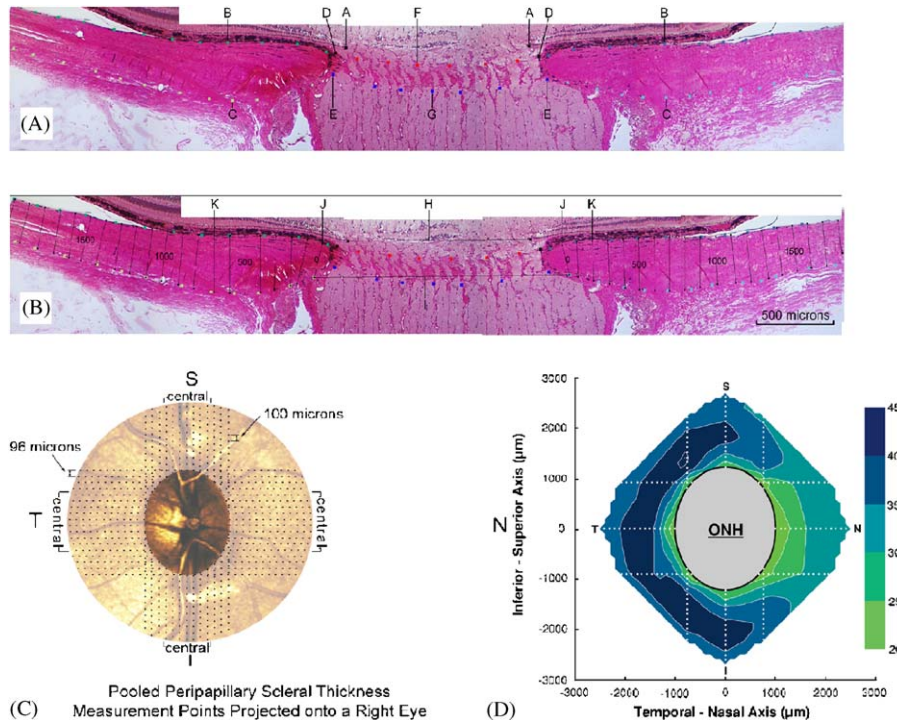


Fig. 24. Peripapillary scleral thickness measurements in normal monkey eyes. (A) Identifying landmarks within a representative serial sagittal section. (B) Semi-automated measurement of scleral thickness at 100-μm intervals from the posterior scleral canal opening (PSCO). (C) Actual location of pooled peripapillary scleral thickness measurements within the serial sagittal sections. (D) Interpolated peripapillary scleral thickness values based on data from nine eyes. Adapted with permission from Downs, J.C., Blidner, R.A., Bellezza, A.J., Thompson, H.W., Hart, R.T., Burgoyne, C.F., 2002. Peripapillary scleral thickness in perfusion-fixed normal monkey eyes. *Invest. Ophthalmol. Vis. Sci.* 43, 2229–2235.

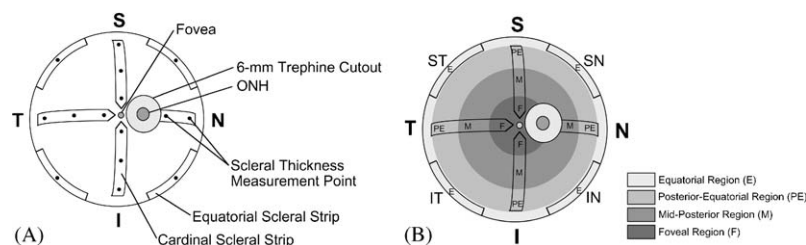


Fig. 25. Posterior scleral thickness measurements. (A) The 15 measurement points within each eye. (B) Regionalization into foveal, mid-posterior, posterior-equatorial, and equatorial points. Reprinted with permission from Downs, J.C., Ensor, M.E., Bellezza, A.J., Thompson, H.W., Hart, R.T., Burgoyne, C.F., 2001. Posterior scleral thickness in perfusion-fixed normal and early-glaucoma monkey eyes. *Invest. Ophthalmol. Vis. Sci.* 42, 3202–3208.

some forms of the neuropathy whereas pressure mechanisms predominate in others. Our conceptual framework suggests that the clinical appearance of the ONH reflects the relative involvement of the neural and/or connective tissues, not the mechanism of insult. Thus, although exceptions exist (Quigley and Anderson, 1977a; Trobe et al., 1980), it predicts that damage to the connective tissues that causes a progressive loss of their ability to resist force is usually required for the ONH surface to deform (bow back and excavate) and assume a glaucomatous appearance. Conversely, as in other forms of damage to the axons with secondary loss of prelaminar capillaries, including ischemic, compres-

sive, and inflammatory insults to the nerve head, nerve, or chiasm, IOP-related (or non-IOP-related) damage to the axons alone, by whatever mechanism, would be expected to cause pallor of the ONH, but not true glaucomatous excavation (Quigley and Anderson, 1977b; Quigley et al., 1982b, 1984).

Two important corollaries evolve from this observation. First, IOP-related damage to the load-bearing connective tissues of the ONH should be central to the pathophysiology of most progressive optic neuropathies that appear excavated, regardless of the IOP at which that optic neuropathy occurs (Fig. 4) or the principal underlying cause. Thus, in the clinical setting of

progressive ONH excavation by whatever mechanism, consideration should be given to aggressively lowering IOP so as to reduce IOP-related strain within the connective tissues, and an appropriate work-up for non-IOP-related causes should then be performed if the clinical indications for a non-IOP-related cause warrant further evaluation.

Second, glaucomatous cupping may not be the only form of IOP-related optic neuropathy. Because IOP-related strain can adversely influence volume flow and diffusion at all levels of IOP, it may also contribute to some progressive optic neuropathies principally characterized by pallor. Our model predicts the existence of a subset of patients with elevated IOP who develop visual field loss that is accompanied by a pale but not excavated ONH (Fig. 32). In these patients, our model

suggests that elevated IOP generates a level of IOP-related strain that is within the elastic limits of the individual ONH's load-bearing connective tissues; thus no excavation occurs. However, the IOP-related strain still causes axonal compromise by one or more of the mechanisms outlined above. Less obviously, our paradigm suggests that when IOP is not measurably elevated and ONH pallor exceeds excavation, an extensive work-up for non-IOP-related causes should be performed and IOP lowering might be additionally considered as a means of increasing perfusion to the axons within the prelaminar, laminar, and retrolaminar regions of the ONH.

10.3. Implication 3

There is a logic for lowering IOP to both protect the load-bearing connective tissues and enhance ONH perfusion. The logic for lowering IOP to protect the connective tissues is apparent from the extensive discussions above and from Fig. 7. However, our paradigm suggests that, at all ages and at all stages of IOP-related damage, delivery of nutrients to the ONH axons can be adversely affected both by retrobulbar determinants of ONH perfusion (e.g., blood pressure and vasospasm) and by the additional negative effects of elevated levels of IOP-related strain within the ONH connective tissues (Fig. 6).

10.4. Implication 4

Progression of ONH damage following IOP lowering is still most likely the result of inadequate IOP lowering. The continued damage to some ONHs following some degree of IOP lowering is often cited as evidence that factors other than IOP are always most important in this

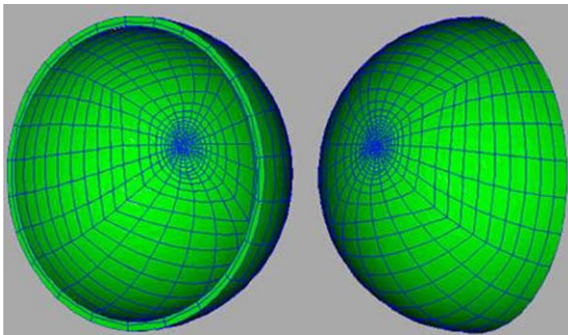


Fig. 26. Representative posterior scleral shell geometry. The actual computational model will be three elements thick (as opposed to the two-element-thick sclera shown above) and will possess a pooled surface curvature and thickness based on histologic measurements of a group of normal and early glaucoma monkey eyes (see Figs. 23–25). Reprinted with permission from Bellezza, A.J., Hart, R.T., Burgoyne, C.F., 2000. The optic nerve head as a biomechanical structure: initial finite element modeling. *Invest. Ophthalmol. Vis. Sci.* 41, 2991–3000.

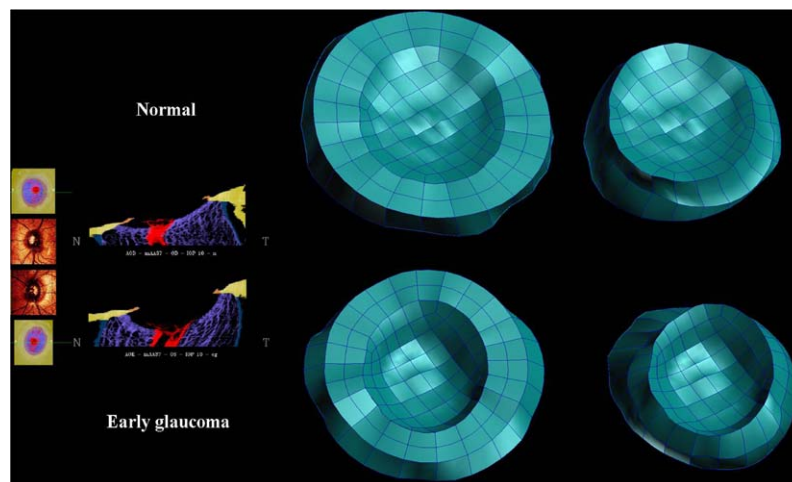


Fig. 27. Representative continuum finite element models for the normal eye (upper) and early glaucoma eye (lower) of an individual monkey. In these continuum models, the histologic boundaries of the principal connective tissue structures have been isolated within the 3D geometries, surfaced, filled, then discretized into relatively large elements as depicted in Fig. 21. The models on the left contain two elements of peripapillary sclera. The models on the right contain just those elements within the scleral canal of each eye.

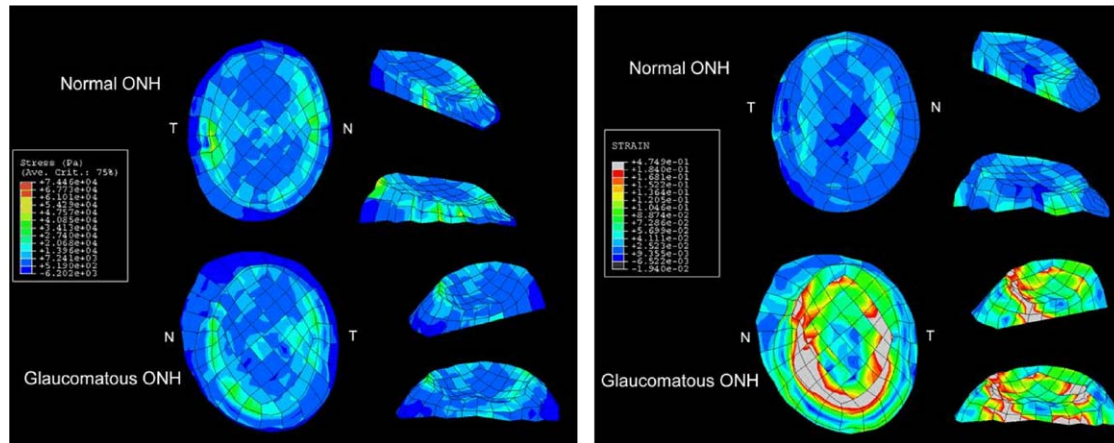


Fig. 28. Representative continuum model output: stress (left window) and strain (right window) within each continuum element of the normal eye (upper panel) and early glaucoma eye (lower panel) of one monkey following acute IOP elevation from 10 to 30 mm Hg. In this exercise, the two eyes of the monkey are treated similarly; however, the material properties and deformation that are used as input to the models are the pooled data for normal and early glaucoma monkey eyes previously outlined. In this case, the material properties of each continuum element of the normal eye are kept the same (treated isotropically); however, the magnitude of this shared value is altered until the laminar deformation within the model after acute IOP elevation from 10 to 30 mm Hg is similar to the overall average values within our previous experiments involving normal and early glaucoma eyes (Table 3). The output of the models are the stresses (left window) and strains (right window) within each continuum element under these conditions. In each window, all of the elements in the canal are depicted on the left, and various sections through these elements are depicted on the right.

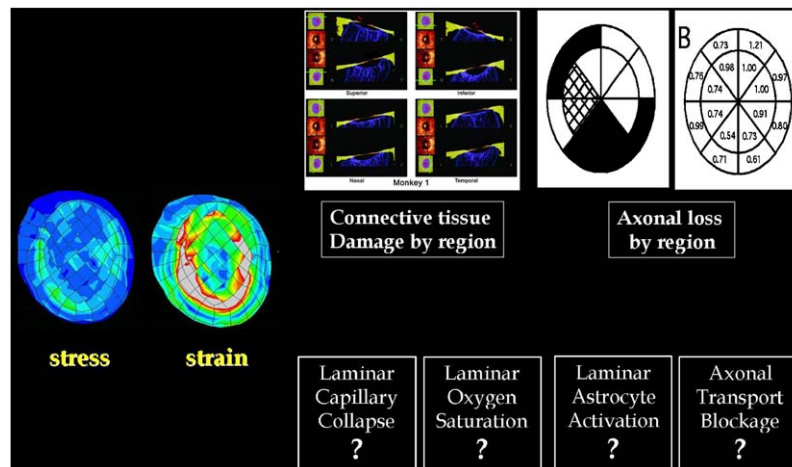


Fig. 29. IOP-related stress and strain underlie the multiple pathophysiologicals of glaucomatous damage. Our hypothesis is that, by definition, a glaucomatous optic neuropathy is one in which both the neural and connective tissues are damaged in relationship to the distribution of stress and strain within the load-bearing connective tissues. We are building the ability to test the hypothesis that the location of greatest stress and strain predicts the location of connective tissue damage and axonal loss within individual monkey optic nerve heads. In the future, we will test the hypothesis that laminar capillary patency, oxygen saturation, astrocyte activation, and axonal transport blockage will also occur within those regions in which the laminar connective and neural tissues experience the greatest levels of stress and strain.

form of progression. Our paradigm is compatible with the notion that non-IOP-related factors such as nocturnal hypotension or immune-mediated insults may predominate in some eyes. However, it also suggests that (1) because we lack true 24-h characterization and control, the actual level of IOP lowering is never known and (2) it is to be expected that severely compromised ONH connective tissues are more susceptible and, therefore, may require very low levels of IOP to remain stable (Ernest and Potts, 1968).

10.5. Implication 5

If an ocular hypertensive monkey eye that has moderate IOP elevation (IOP in the 20s and 30s) is followed until the first detectable change in the CSLT appearance, which usually occurs in about 4–8 weeks, and then is perfusion fixed with both eyes at an IOP of 10 mm Hg, marked deformation of the scleral canal and lamina is seen in the eye with early experimental glaucoma, compared with the contralateral normal eye

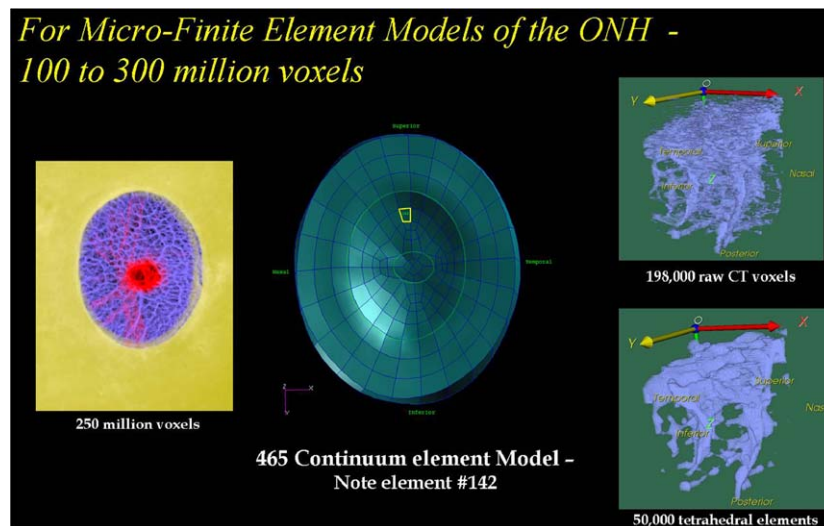


Fig. 30. Our strategy for micro finite element modeling. Continuum modeling simplifies the 250 million voxel 3D geometry of the peripapillary sclera into a computational model of 465 continuum elements. In micro finite element modeling, the laminar beams within an individual continuum element will be surfaced and discretized into elements, with the remaining continuum model being used to establish their boundary conditions. Here the 195,000 connective tissue voxels of continuum element 142 are visualized in the upper right panel and are surfaced and discretized into a model of 55,000 tetrahedral finite elements (lower right panel).

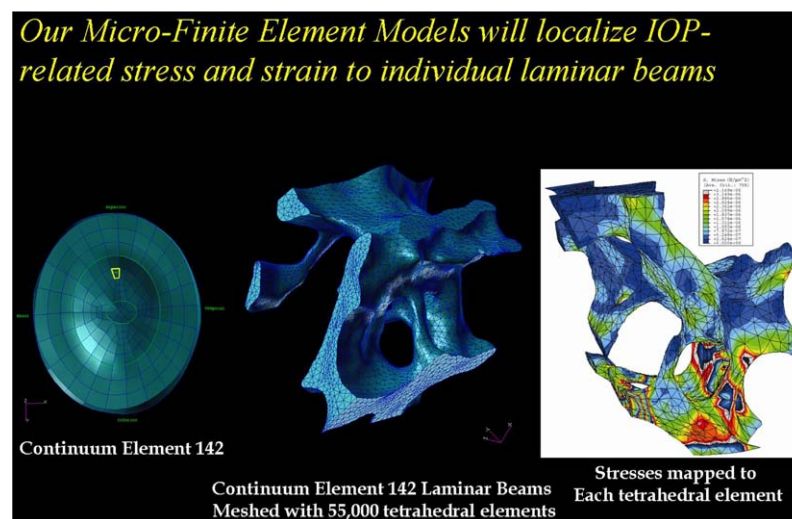


Fig. 31. Focal stresses within the micro FEM of the individual laminar beams of a representative continuum finite element. Here stresses and strains are mapped to the individual laminar beams of continuum element 142. In our initial micro FEMs, stresses within the individual laminar beams have been as much as 33 times greater than the continuum stress of the parent continuum element.

(Fig. 17) (Downs et al., 2003a). This is at least the engineering equivalent to “yield” and may in some regions equate to catastrophic failure of individual laminar beams. To date, this has been studied for the most part in young adult monkeys (equivalent to 18–35 years of age in humans), which may be substantially more susceptible than most humans, but we predict that the onset of CSLT-detected ONH surface change in humans will have the same underlying connective tissue correlates. Thus, from a purely

engineering standpoint, waiting for the onset of CSLT-detected surface change to lower IOP means accepting a level of damage to the underlying connective tissues that will increase the susceptibility of the remaining connective tissues to further damage. However, this degree of connective tissue damage may not increase susceptibility to a level that is clinically important. In which human eyes or in how many eyes this is a clinically important increase in susceptibility remains to be determined.

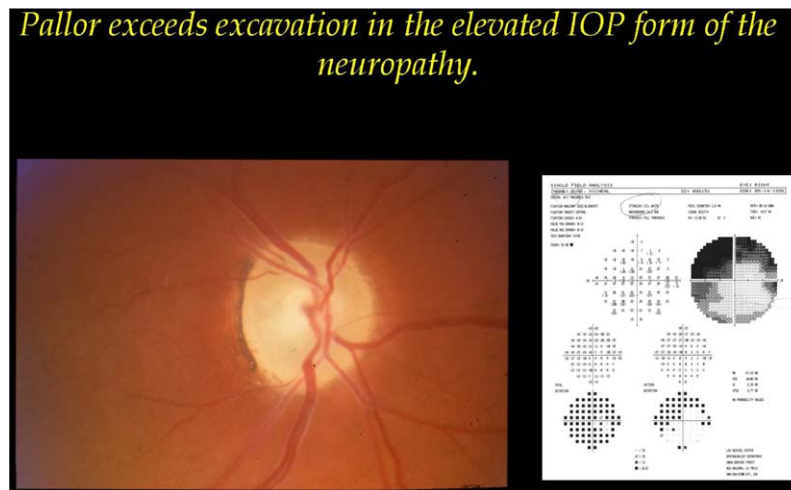


Fig. 32. Pallor exceeds excavation in the elevated IOP form of the neuropathy, as shown in the ONH photo (left) and 10-2 visual field (right) of the right eye of a 42-year-old patient with a long history of chronically elevated IOP. In this particular eye, there had been substantial reversal of cupping following a successful glaucoma filtration surgery. Although the amount of remaining rim tissue is substantial, both pallor and visual field loss are profound.

11. Summary

The principal appeal of modeling the ONH as a biomechanical structure is that it provides a conceptual framework that explains several heretofore confusing aspects of the disease. First, our theory of a predictable cascade of IOP-induced mechanical failure within the load-bearing connective tissues provides the first testable hypotheses regarding the specific events underlying the onset and progression of glaucomatous excavation.

Second, the biomechanical notion that IOP-related stress within the sclera is magnified within and immediately around the scleral canal (Greene, 1980; Bellezza et al., 2000), and that this elevated level of stress centrally determines the physiologic and pathophysiologic milieu, provides a new logic for understanding how a glaucomatous form of damage to the ONH can occur at all levels of IOP depending on the unique susceptibility of an individual ONH.

Third, the notions of elevated levels of IOP-related stress within the peripapillary sclera and IOP-induced basement membrane and connective tissue changes within the lamellar trabeculae allow our model to hypothesize a series of interactions between IOP, IOP-related stress, IOP-related strain, ONH volume blood flow, and the delivery of blood-borne nutrients to the axons within the lamina cribrosa. This hypothesis suggests that vascular mechanisms of damage to the ONH are not necessarily IOP-independent and proposes a logic for understanding the complicated interaction between these three important factors—IOP, volume blood flow, and nutrient delivery—within the tissues of the ONH.

Fourth, the concept of mechanical failure suggests a biomechanical explanation for the clinical notion that

progression of ONH damage after lowering of IOP is most likely caused by inadequate pressure reduction. Our model suggests that diurnally (24-h) confirmed levels of low IOP may be necessary in some eyes because once the load-bearing connective tissues are focally damaged, stresses are higher within the remaining tissues bearing load.

Fifth, our paradigm suggests a logic for characterizing the multiple effects of IOP-related strain on the individual tissues of the visual system (Fig. 2). Although the precise steps within each staging pathway will be unique to each tissue, the general progression from molecular changes, to histologic evidence of alteration and damage, to histologic evidence of repair and/or regeneration should have equivalent manifestations in all of the individually affected tissues.

12. Future directions

Our goal is to build a science that predicts the susceptibility of the *connective tissues* of an individual ONH to a given level of IOP. Eventually, our models will identify the principal determinants of the level of IOP-related stress for a given level of IOP. Then, if clinical measures of these parameters can be developed, which may eventually be achieved with micro magnetic resonance imaging (Sadun et al., 2002), finite element modeling of the ONH connective tissues within individual eyes may become a standard clinical test to determine target IOPs for eyes that are at risk for damage.

We are beginning to extend our studies to the associated susceptibilities of the *astrocytes* and *retinal ganglion cell axons* within the ONH. We are building cell

and tissue culture systems to characterize the short- and long-term molecular response of astrocytes to stress and strain. Eventually, we hope to use probes for astrocyte activation and axonal distress in order to map the locations of these phenomena within our 3D connective tissue reconstructions relative to the levels of connective tissue stress and strain at varying stages of glaucomatous damage.

We will soon begin studies designed to characterize the biochemical and histologic alterations that underlie changes in the material properties of the posterior sclera in early glaucoma. Our ultimate goal will be to describe the histologic and ultrastructural changes of mechanical failure within the peripapillary sclera, laminar trabeculae, and scleral canal wall in early, moderate, and severe glaucomatous damage.

Finally, our 3D reconstructions contain the posterior ciliary arteries, their principal branches to the pre-laminar, laminar, and post-laminar ONH, and the capillaries within each laminar beam. We will eventually model the effects of IOP-related stress and strain, aging, and the stage of glaucomatous damage on volume flow within this blood supply and the diffusion of nutrients from the laminar capillaries through the laminar extracellular matrix, astrocyte basement membrane, and astrocyte cell processes into the center of the axon bundles.

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